

Induced-Gravity Cosmology: Inflation, Dark Energy, and Dark Matter from a Common Scalar Origin

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Abstract

We construct a scalar-tensor cosmology in pure induced gravity: a real scalar Φ with $\mathbb{Z}_2$ potential $V_J(\Phi) = \frac{\lambda_0}{4}(\Phi^2 - \Phi_0^2)^2 + V_c$ and coupling $\xi\Phi^2 R$ replaces the Einstein–Hilbert term, generating M_{Pl} via $\langle\Phi\rangle = M_{\text{Pl}}/\sqrt{\xi}$. Conformal transformation yields a Starobinsky-type slow-roll plateau. With locked $N \equiv \ln(a_{\text{end}}/a_*)$ and exact potential slow-roll, $\xi = 11.1$ gives $N = 50 \Rightarrow \lambda_0 \simeq 6.70 \times 10^{-8}$, $r \simeq 0.00425$, $n_s \simeq 0.962$; anomaly $T_{\text{reh}} \simeq 10^9$ GeV maps to $N \simeq 50.7$. Planck 2σ selects $N \approx 45\text{--}56$ ($r \leq 0.0052$; capped at $N_{\text{max}} \simeq 55.6$ by instantaneous reheating); LiteBIRD $r > 0.01$ would exclude the model class. A Dirac fermion ψ with $-g\Phi\bar{\psi}\psi$ is a cold-dark-matter *candidate*: $m_\psi = g\Phi_0$ shares the VEV; stability relies on an assumed anomaly-free discrete $\mathbb{Z}_2^\psi$ plus the conformally vanishing Yukawa vertex. Matching Ω_{DM} with the exact mode-equation spectrum on the exact homogeneous background (transition and oscillating-condensate era) selects $g \simeq 1.5 \times 10^{-7}$ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV $\simeq 6.6 \times 10^{-3} H_{\text{inf}}$), cold kinematics ($\lambda_{\text{fs}} \sim 10^{-19}$ Mpc); absolute normalization is closed at the homogeneous level (convergence/unitarity/conformal battery), with transition-shape systematic ($\times 1.6$) and a 3D preheating cross-check remaining. The potential minimum gives a frozen cosmological constant $w = -1$ (residuals $\leq 10^{-10}$); dark energy is calibrated, not predicted (V_c fixed by $\Omega_\Lambda = 0.683$, 10^{110} hierarchy unexplained); confirmed DESI DR2 preference for $w_0 \neq -1$ would exclude it and ΛCDM alike. Tree-level $\chi \rightarrow h\bar{h}$ and $\chi \rightarrow \psi\bar{\psi}$ close; reheating proceeds via conformal anomaly. SM embedding is RG-stable ($\Delta\xi \leq 10^{-6}$); $w_0 = -1$ is shared with ΛCDM ; the sharp falsification handles remain inflationary.

Keywords: induced gravity, scalar–tensor cosmology, Starobinsky plateau, gravitational particle production, superheavy dark matter, cosmological constant

1 Introduction

Modern Λ CDM cosmology achieves remarkable observational success, yet the physical origins of its key components—inflation, dark matter, and dark energy—remain unresolved. The Einstein–Hilbert term is treated as fundamental, and the three sectors are described by independent mechanisms with no shared microphysics.

Induced gravity [1, 2] offers an elegant conceptual simplification: the Planck mass is not fundamental but emerges from the vacuum expectation value of a real scalar field Φ . The Einstein–Hilbert term $M_{\text{Pl}}^2 R/2$ is replaced by $\xi \Phi^2 R/2$; without field condensation, there is no spacetime gravity. At low energies, General Relativity is recovered as $\Phi \rightarrow M_{\text{Pl}}/\sqrt{\xi}$.

The Einstein-frame potential is shown in Fig. 1. The domain-wall / EFT-boundary structure at $\Phi = 0$ is illustrated in Fig. 3. At large field values, the non-minimal coupling produces, after conformal transformation to the Einstein frame, a Starobinsky-type slow-roll plateau [3, 4]. The plateau geometry follows unavoidably from the same coupling that generates M_{Pl} .

Once this step is taken, a single field Φ with a single $\mathbb{Z}_2$ -symmetric potential $V_J(\Phi)$ can simultaneously account for:

- **The Planck scale:** $\langle \Phi \rangle = M_{\text{Pl}}/\sqrt{\xi}$ gives the observed gravitational coupling.
- **Inflation:** at $\Phi \gg \Phi_0$, conformal transformation yields a Starobinsky-type plateau. Attractor forms $n_s \simeq 1 - 2/N$ and $r \simeq 8/(\beta^2 N^2)$ are analytic guides; all numerical values in this paper use the locked cosmological definition $N = \ln(a_{\text{end}}/a_*)$ with exact potential slow-roll (Sec. III), which at $N = 50$ gives $r \approx 0.00425$ rather than the attractor 0.00487.
- **Dark matter:** a Dirac fermion ψ , with mass $m_\psi = g\Phi_0$ set by the same VEV, is gravitationally produced at the end of inflation and is stable under the discrete $\mathbb{Z}_2^\psi$ symmetry together with the conformally vanishing $\chi\psi\bar{\psi}$ vertex; the condensate itself decays to complete reheating. The relic abundance is fixed at the homogeneous level by the exact mode-equation computation of Sec. V (Appendix D0), with a 3D preheating simulation remaining as the non-perturbative cross-check.
- **Dark energy:** the residual vacuum energy V_c at $\Phi = \Phi_0$ gives $w = -1$ (up to residual-condensate corrections; App. D5), *calibrated empirically* to Ω_Λ — not a dynamical prediction of residual inflaton motion.
- **Fifth-force screening:** the conformal identity $\Omega = \Phi/\Phi_0$ (forced by $F = \xi\Phi^2$) decouples the scalar from the dark-matter fermion ψ at tree level; for Standard Model matter, the scalar’s large mass $m_\chi \sim 10^{13}$ GeV (Yukawa range $\sim 10^{-27}$ cm) suppresses fifth forces, automatically satisfying solar-system bounds.
- **$\mathbb{Z}_2$ domain wall:** the surface $\Phi = 0$ is a classical EFT boundary where $F(0) = F'(0) = 0$ renders the Israel junction conditions inconsistent.

Relation to existing work. Induced gravity was originally proposed as an alternative to the Einstein–Hilbert action [1, 2] and later recognized as a natural framework for inflation [5, 6]. Recent work has explored induced-gravity inflation with various potentials [7, 8], while the non-minimal coupling $\xi\Phi^2 R$ is shared with Higgs inflation [3] (though at $\xi \sim 10^4$ rather than $\xi \sim 11.1$). R^2 (Starobinsky) inflation [4] produces

identical $n_s(N)$ and $r(N)$ functional forms but lacks the unified dark-energy/dark-matter embedding. The idea of unifying inflation and dark energy through a single scalar potential has been explored in quintessential inflation [9, 10], typically with disparate scales bridged by tracker or scaling solutions. The present work differs in three respects: (a) pure induced gravity (no independent $M_{\text{Pl}}^2 R/2$ term) with the $\mathbb{Z}_2$ -symmetric double-well potential simultaneously provides the inflationary plateau, the dark-energy vacuum energy, and the dark-matter mass scale from a single VEV; (b) the conformal identity $\Omega = \Phi/\Phi_0$ —a structural consequence of $F = \xi\Phi^2$ —closes the perturbative Yukawa reheating channel and decouples the dark-matter fermion at tree level, making gravitational particle production the sole DM source; (c) the $\chi \rightarrow h\bar{h}$ tree-level decay is structurally closed in induced gravity (in contrast to R^2 inflation), making the conformal-anomaly decay the dominant reheating channel. These features are not imposed by hand but follow unavoidably from the induced-gravity coupling. Two consequences deserve emphasis. First, since the $n_s(N)$ and $r(N)$ functions are numerically identical to those of R^2 inflation, the inflationary sector alone does *not* discriminate this model from Starobinsky inflation at fixed N ; what differs is the input budget, because the same VEV fixes both the Planck mass and the dark-matter mass through the exact relation $m_\psi/H_{\text{inf}} = (g/\sqrt{\xi})(M_{\text{Pl}}/H_{\text{inf}})$. The model’s only non-degenerate observational connection is therefore the dark-matter sector, which is precisely the sector whose normalization is now fixed at the homogeneous level (Sec. V). Second, the gravitational-production channel places the fermion in the wimpzilla class [28–30], but with a quantitatively different outcome: in the standard wimpzilla treatment the relic abundance is exponentially sensitive to m_ψ/H_{inf} and is matched near $m_\psi \sim H_{\text{inf}}$, whereas the exact mode-equation spectrum across the end-of-inflation transition is a power law (Appendix D0), which moves the matching solution down to $m_\psi \simeq 6.6 \times 10^{-3} H_{\text{inf}}$ and makes the abundance scale with the reheating temperature rather than with g .

This paper constructs this unified framework. Section II lays the theoretical foundation. Sections III–IV cover inflation and reheating. Section V treats the gravitationally produced fermion ψ as the dark matter candidate. Section VI derives the frozen Λ CDM dark energy and bounds the residual motion of the field, including its drift of the fundamental constants. Section VII embeds the Standard Model and verifies RG stability. Section VIII analyzes the $\mathbb{Z}_2$ domain wall as a classical EFT boundary. Section IX synthesizes the unified picture. Section X collects falsifiable predictions. Section XI enumerates open problems.

We define the coupling function $F(\Phi) \equiv \xi\Phi^2$, central to all metric variations and junction conditions. Matter fields (fermions ψ) couple through Yukawa interactions $\mathcal{L}_Y = -g\Phi\bar{\psi}\psi$, with $m_\psi = g|\Phi|$. In this paper, “common origin” refers to the fact that all relevant mass scales—the Planck mass M_{Pl} , the dark-matter fermion mass $m_\psi = g\Phi_0$, and the overall scale of $V_J(\Phi)$ —are set by a single vacuum expectation value $\langle\Phi\rangle = \Phi_0$; the dark matter is a fermionic partner whose stability is guaranteed by the same UV gauge structure (the discrete $\mathbb{Z}_2^\psi$ symmetry). We do not claim a microphysical derivation of the dark-energy scale V_c , which is calibrated empirically.

2 Theoretical foundation

2.1 Action and field equations

We adopt the $(-, +, +, +)$ metric signature and the Landau–Lifshitz curvature conventions ($R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \dots$, $R_{\mu\nu} = R^\rho_{\mu\rho\nu}$, $R = g^{\mu\nu} R_{\mu\nu}$).

The Jordan-frame action is

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} \xi \Phi^2 R - \frac{1}{2} g^{\mu\nu} \partial_\mu \Phi \partial_\nu \Phi - V(\Phi) \right] + S_{\text{matter}}, \quad (1)$$

with the symmetry-breaking potential

$$V(\Phi) = \frac{\lambda_0}{4} (\Phi^2 - \Phi_0^2)^2 + V_c, \quad \Phi_0 = \frac{M_{\text{Pl}}}{\sqrt{\xi}}, \quad (2)$$

where V_c is a vacuum-energy constant, $M_{\text{Pl}} \approx 2.435 \times 10^{18}$ GeV, and λ_0 is the self-coupling. The $\mathbb{Z}_2$ symmetry $\Phi \rightarrow -\Phi$ produces degenerate vacua $\pm\Phi_0$ separated by a potential barrier at $\Phi = 0$.

Variation with respect to $g_{\mu\nu}$ yields the modified Einstein equation

$$FG_{\mu\nu} = T_{\mu\nu}^{\text{matter}} + T_{\mu\nu}^{(\Phi)} + \nabla_\mu \nabla_\nu F - g_{\mu\nu} \square F, \quad (3)$$

with canonical scalar energy-momentum tensor $T_{\mu\nu}^{(\Phi)} \equiv \partial_\mu \Phi \partial_\nu \Phi - \frac{1}{2} g_{\mu\nu} (\partial\Phi)^2 - g_{\mu\nu} V(\Phi)$. The scalar equation of motion is

$$\square\Phi - V'(\Phi) + \xi\Phi R = 0. \quad (4)$$

We distinguish three field variables: the Jordan-frame field Φ ; the logarithmic parameter $\varphi \equiv \ln(\Phi/\Phi_0)$ with conformal factor $\Omega^2 = e^{2\varphi}$; and the Einstein-frame canonical field χ ($d\chi/d\varphi = M_{\text{Pl}}\sqrt{6 + 1/\xi}$).

2.2 $\mathbb{Z}_2$ symmetry and double vacua

The action is invariant under $\Phi \rightarrow -\Phi$; extrema satisfy $\Phi \in \{0, \pm\Phi_0\}$. The domain-wall tension is

$$\sigma_{\text{wall}} = \int_{-\Phi_0}^{+\Phi_0} d\Phi \sqrt{2[V(\Phi) - V(\Phi_0)]} = \frac{4}{3} \sqrt{\frac{\lambda_0}{2}} \Phi_0^3 > 0. \quad (5)$$

Cosmological domain-wall networks are avoided through initial-condition locking during inflation and inflationary dilution: since $m_\Phi = \sqrt{2\lambda_0}\Phi_0 \simeq 2.7 \times 10^{14}$ GeV $\gg H_{\text{inf}} \simeq 1.64 \times 10^{13}$ GeV, the field relaxes to one vacuum branch ($+\Phi_0$ or $-\Phi_0$) within a few m_Φ^{-1} , well before the CMB scales exit the horizon; inflation then stretches this choice beyond the present horizon, so the standard post-inflationary Kibble wall-formation argument does not apply. The relevant suppression is in *number density*, not energy density: the number of walls per comoving volume is diluted by $n_{\text{wall}} \propto a^{-3} \sim e^{-3N}$, giving a probability $\sim e^{-3N} \sim 10^{-66}$ that a single wall intersects

our observable patch. (For completeness: the wall tension $\sigma_{\text{wall}} \sim (4/3)\sqrt{\lambda_0/2}\Phi_0^3 \sim 10^{50} \text{ GeV}^3$ vastly exceeds the CMB bound $\sigma \leq (1 \text{ MeV})^3$, so any wall in our patch would be catastrophic—but the number-density suppression by inflation makes this exponentially unlikely.) Bias-potential elimination remains an open problem.

2.3 Unified mass scales from the scalar VEV

The key unification mechanism is that all mass scales originate from the VEV of Φ :

a. Planck mass.

The effective gravitational constant is

$$G_{\text{eff}} = \frac{1}{8\pi\xi\Phi^2} \implies G_N = \frac{1}{8\pi\xi\Phi_0^2}. \quad (6)$$

At $\Phi = \Phi_0$, $G_{\text{eff}} = G_N$ matches Newton’s constant. The active order parameter is $\mathcal{A} \equiv (\Phi/\Phi_0)^2$, with $G_{\text{eff}} = G_N/\mathcal{A}$.

On the relation $M_{\text{Pl}}^2 = \xi\Phi_0^2$.

In induced gravity this is a definition, not a tuning: the Planck mass is not an independent input but is generated by the VEV, $M_{\text{Pl}}^2 \equiv \xi\langle\Phi\rangle^2$. The potential $V(\Phi) = \frac{\lambda_0}{4}(\Phi^2 - \Phi_0^2)^2 + V_c$ has its extrema at $\Phi = \pm\Phi_0$ exactly (the constant V_c does not shift the minimum), so the VEV coincides with Φ_0 as a consequence of the potential’s construction. A quantum shift of the VEV would rescale M_{Pl} proportionally, leaving the dimensionless identity $\Omega = \Phi/\Phi_0$ (Eq. (10))—which depends only on the coupling form $F = \xi\Phi^2$, not on the numerical value of Φ_0 —intact. The conformal decoupling derived in Sec. II E is therefore a structural property of the induced-gravity coupling, not a fine-tuned coincidence.

b. Fermion masses.

Fermions ψ couple through the Yukawa interaction $\mathcal{L}_Y = -g\Phi\bar{\psi}\psi$, giving $m_\psi = g|\Phi|$. At the minimum, $m_\psi^{(0)} = g\Phi_0$. A full Standard Model embedding (including Higgs-mediated masses and RG stability) is constructed in Sec. VII.

c. Weak scale (optional Higgs portal).

If a Higgs portal $\lambda_{\Phi H}\Phi^2|H|^2$ is included, the Higgs VEV becomes

$$v_H^2 = \frac{\mu_H^2 - \lambda_{\Phi H}\Phi_0^2}{2\lambda_H}, \quad (7)$$

linking the electroweak scale to Φ_0 and hence to M_{Pl} . This preserves the $\mathbb{Z}_2$ symmetry (Φ^2 is even) and is technically natural [11].

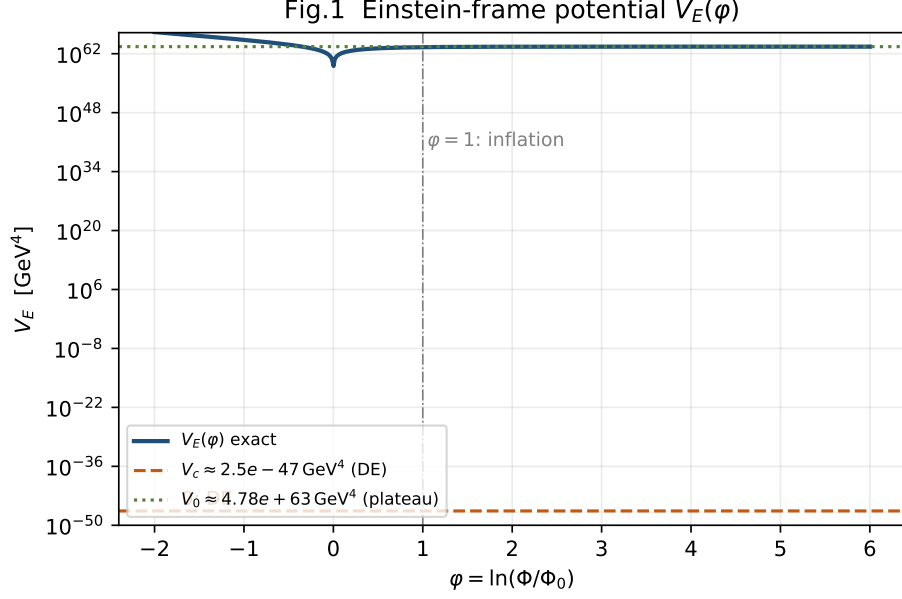


Fig. 1 Exact Einstein-frame potential $V_E(\varphi)$ for locked $N = 50$, $\xi = 11.1$ ($\lambda_0 \simeq 6.70 \times 10^{-8}$ from exact slow-roll A_s). The flat plateau at $\varphi \gg 1$ ($\sim V_0$) drives slow-roll inflation; the minimum at $\varphi \simeq 0$ is set by the calibrated V_c . The hierarchy $V_0/V_c \sim 10^{110}$ is spanned by the conformal factor $e^{-4\varphi}$ but not dynamically explained. Regenerated by `scripts/fig1.einstein-potential.py`.

2.4 Conformal transformation to Einstein frame

Under $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu} = e^{2\varphi} g_{\mu\nu}$ with $\Omega^2 \equiv \xi \Phi^2 / M_{\text{Pl}}^2$, the exact Einstein-frame potential is

$$V_E(\varphi) = \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2} (1 - e^{-2\varphi})^2 + V_c e^{-4\varphi}. \quad (8)$$

Two limits: $\varphi \gg 1$ gives $V_E \rightarrow \lambda_0 M_{\text{Pl}}^4 / (4\xi^2) \equiv V_0$ (inflationary plateau); $\varphi \rightarrow 0$ gives $V_E \rightarrow V_c$ (frozen cosmological constant).

With the canonical field χ , the potential assumes the Starobinsky form [4]

$$\frac{d\chi}{d\varphi} = M_{\text{Pl}} \sqrt{6 + \frac{1}{\xi}}, \quad V_E(\chi) = \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2} \left(1 - e^{-\beta\chi/M_{\text{Pl}}}\right)^2, \quad \beta \equiv \frac{2}{\sqrt{6 + 1/\xi}}, \quad (9)$$

where $\beta \simeq 0.810$ for $\xi \sim 11.1$. The V_c term is negligible on the plateau but determines the late-time dark-energy density.

2.5 Conformal decoupling and automatic fifth-force screening

A single exact identity of this model decouples the scalar field from the dark-matter fermion ψ , while the scalar's large mass screens the fifth force for Standard Model

matter. Because $F(\Phi) = \xi\Phi^2$, the conformal factor is exactly proportional to the field:

$$\Omega = \frac{\sqrt{\xi}\Phi}{M_{\text{Pl}}} = \frac{\Phi}{\Phi_0} \implies \Phi = \Phi_0 \Omega, \quad (\text{exact, all } \Phi). \quad (10)$$

This proportionality is specific to the induced-gravity coupling; a generic $F(\Phi)$ would not enjoy it.

We distinguish two classes of fermions. For fermions whose mass arises entirely from the direct Yukawa coupling $-g\Phi\bar{\psi}\psi$, the conformal identity produces exact decoupling. Under $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ and $\tilde{\psi} = \Omega^{-3/2}\psi$ (equivalently $\psi = \Omega^{3/2}\tilde{\psi}$), the Yukawa term transforms as

$$-g\Phi\bar{\psi}\psi\sqrt{-g} \longrightarrow -g\Phi_0\bar{\tilde{\psi}}\tilde{\psi}\sqrt{-\tilde{g}} : \quad (11)$$

the Ω factors cancel exactly ($\Omega \cdot \Omega^3 \cdot \Omega^{-4} = 1$), giving an *exact*, χ -independent Einstein-frame fermion mass $m_E = g\Phi_0$. The tree-level $\chi\bar{\psi}\psi$ coupling therefore vanishes identically, so the PPN parameter $\gamma = 1$ holds exactly within the ψ sector. At loop level, graviton exchange can induce an effective $\chi\bar{\psi}\psi$ vertex of parametric size $\leq m_\psi m_\chi / M_{\text{Pl}}^2 \sim 10^{-11}g$ —utterly negligible for reheating, dark matter production, and fifth-force tests, so the tree-level vanishing is the operative feature. This ψ -sector decoupling is observationally inert for solar-system tests, which probe Standard Model matter (no ψ sources are present in the solar system); it is instead decisive for the reheating and dark-matter analysis (Secs. IV–V).

Standard-model fermions obtain their mass through the Higgs mechanism, so the conformal identity does *not* cancel their χ coupling; a residual coupling $\sim m_f/M_{\text{Pl}}$ remains (treated separately in Sec. VII). The solar-system fifth-force bound is satisfied by the dominant screening mechanism, namely the scalar’s large canonical mass near the minimum,

$$m_\chi^2 = \frac{V_E''(0)}{M_{\text{Pl}}^2(6 + 1/\xi)} \approx \frac{2\lambda_0 M_{\text{Pl}}^2}{\xi^2(6 + 1/\xi)} \approx (3.25 \times 10^{13} \text{ GeV})^2, \quad (12)$$

(at locked $N = 50$, $\xi = 11.1$; $3.19 \times 10^{13} \text{ GeV}$ at $N = 51$), which gives a Yukawa range $m_\chi^{-1} \sim 10^{-27} \text{ cm}$, exponentially suppressing any residual interaction at macroscopic distances and satisfying the Cassini bound $|\gamma - 1| < 2.3 \times 10^{-5}$ [12]. This short-range suppression is the same mechanism operative in any heavy-scalar-tensor theory; it is distinct from a chameleon field-environment mechanism, which is not invoked here.

Key differences from standard Higgs inflation.

While the plateau geometry is shared: (a) $\xi \sim 11.1$ is two orders below $\xi \sim 10^4$, yielding a different unitarity regime; (b) M_{Pl} is dynamically generated via induced gravity; (c) the same $V_J(\Phi)$ supplies both the inflationary plateau and the late-time vacuum constant V_c ; (d) Φ is a gauge singlet, so the large top-Yukawa term that drives the Higgs β_ξ is absent.

2.6 EFT cutoff scale

The status of the cutoff is frame-dependent and, for non-minimal scalar-tensor inflation, *contested*. From the Einstein-frame EFT perspective [3], the action is standard GR plus a canonical scalar; higher operators are suppressed by M_{Pl} and $H_{\text{inf}} \ll M_{\text{Pl}}$ ensures self-consistency. From the Jordan-frame perspective [13, 14], the cutoff is $\Lambda_J \sim M_{\text{Pl}}/\xi \sim 2.2 \times 10^{17} \text{ GeV}$, and the CMB pivot value Φ_{CMB} exceeds it by ~ 30 . Whether the low Jordan-frame cutoff persists as a physical limitation of the Einstein-frame EFT is an open question without community consensus [15]. We adopt the Einstein-frame EFT perspective for computing slow-roll observables (where they are robust) and flag the Jordan-frame cutoff tension as an open issue (Discussion #8).

3 Inflation: Starobinsky-type slow-roll plateau

Conditional statement.

The derivation describes the geometric fact that a slow-roll plateau exists whenever $\chi \gg M_{\text{Pl}}$. The naturalness of this initial condition is not endogenously explained (Discussion #1).

3.1 Energy scale and CMB normalization

Definition of N and CMB normalization.

Throughout this paper the e -fold number is the cosmological quantity

$$N \equiv \ln \left(\frac{a_{\text{end}}}{a_*} \right), \quad k_* = a_* H_*, \quad (13)$$

with pivot $k_* = 0.05 \text{ Mpc}^{-1}$. Post-inflationary matching assumes matter-like ($w = 0$) condensate domination until T_{reh} ,

$$a_{\text{end}} = a_{\text{reh}} \left(\frac{\rho_{\text{reh}}}{\rho_{\text{end}}} \right)^{1/3}, \quad a_{\text{reh}} = a_{\text{eq}} \left(\frac{g_{*s,\text{eq}}}{g_{*s,\text{reh}}} \right)^{1/3} \frac{T_{\text{eq}}}{T_{\text{reh}}}, \quad a_{\text{eq}} = \frac{\Omega_r}{\Omega_m}, \quad (14)$$

with $\Omega_m \simeq 0.315$, $\Omega_r = 9.15 \times 10^{-5}$ (the Planck value, $\Omega_r h^2 = 4.15 \times 10^{-5}$ with $h = 0.674$), $\rho_{\text{reh}} = (\pi^2/30)g_* T_{\text{reh}}^4$, $\rho_{\text{end}} = K_{\text{end}} + V_{\text{end}}$, $g_{*s,\text{reh}} = 106.75$, $g_{*s,\text{eq}} = 3.91$, $g_{*,\text{eq}} = 3.36$ and $T_{\text{eq}} = (30\rho_{\text{eq}}/\pi^2 g_{*,\text{eq}})^{1/4}$. The radiation era is matched by comoving-entropy conservation rather than by the constant- g_* scaling $a \propto \rho^{-1/4}$; the two routes differ by a factor $\simeq 2.5$ in T_{reh}^* , i.e. $\simeq 0.30$ e -folds in N (the combined effect of entropy matching, the $\rho_{\text{end}} = K_{\text{end}} + V_{\text{end}}$ update and the Ω_r update; the g_{*s} dilution $106.75 \rightarrow 3.91$ alone would give $(106.75/3.91)^{1/3} \simeq 3.0$). At a given N , the plateau variable x_* is fixed by the exact potential-slow-roll integral

$$N = \frac{e^{x_*} - x_* - (e^{x_{\text{end}}} - x_{\text{end}})}{2\beta^2}, \quad e^{-x_{\text{end}}} = \frac{1}{1 + \sqrt{2}\beta}, \quad \beta \equiv \frac{2}{\sqrt{6 + 1/\xi}}, \quad (15)$$

and observables are evaluated at that x_* :

$$r = 16\varepsilon_*, \quad n_s = 1 - 6\varepsilon_* + 2\eta_*, \quad A_s = \frac{V_0(1 - e^{-x_*})^2}{24\pi^2 M_{\text{Pl}}^4 \varepsilon_*}, \quad (16)$$

with

$$\varepsilon = \frac{2\beta^2 e^{-2x}}{(1 - e^{-x})^2}, \quad \eta = \frac{2\beta^2 e^{-x}(2e^{-x} - 1)}{(1 - e^{-x})^2}, \quad V_0 = \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2}. \quad (17)$$

Solving $A_s = 2.100 \times 10^{-9}$ at fixed (N, ξ) determines λ_0 uniquely (linear in λ_0 at fixed x_*). Attractor-style closed forms $r \simeq 8/(\beta^2 N^2)$, $n_s \simeq 1 - 2/N$ remain useful approximations but *must not* be mixed with this N definition: at $N = 50$ the exact potential slow-roll gives $r \approx 0.00425$, whereas the closed form yields $r \approx 0.00487$.

Fiducial point.

For $\xi = 11.1$ and $N = 50$ under Eq. (13):

$$\lambda_0 \simeq 6.70 \times 10^{-8}, \quad r \simeq 0.00425, \quad n_s \simeq 0.9616, \quad (18)$$

$$H_{\text{inf}} = \sqrt{\frac{V_0}{3M_{\text{Pl}}^2}} \simeq 1.64 \times 10^{13} \text{ GeV}, \quad U^{1/4} \simeq 8.3 \times 10^{15} \text{ GeV}, \quad m_\chi \simeq 3.25 \times 10^{13} \text{ GeV}, \quad (19)$$

where $m_\chi^2 = 8V_0/(M_{\text{Pl}}^2 \beta_o^2) = 2\lambda_0 M_{\text{Pl}}^2/[\xi^2(6 + 1/\xi)]$ with $\beta_o = \sqrt{6 + 1/\xi}$. The conformal-anomaly reheating estimate $T_{\text{reh}} \sim 10^9 \text{ GeV}$ maps through Eqs. (14)–(15) to $N \simeq 50.7$ (not 50); the corresponding self-consistent temperature for $N = 50$ is $T_{\text{reh}} \simeq 1.1 \times 10^8 \text{ GeV}$. Matching assumptions (k_* , w during reheating, Ω 's) shift these temperatures by $\mathcal{O}(1)$ – $\mathcal{O}(10)$.

Sensitivity table.

Table 1 lists exact slow-roll predictions at $\xi = 11.1$ under the locked N definition. Planck 2σ ($n_s \in [0.9565, 0.9733]$) allows, on n_s alone, $N \approx 45$ – 56 with $r \in [0.0034, 0.0052]$; the tighter 1σ lower edge $n_s \geq 0.9607$ selects $N \geq 49$. The upper edge is not set by n_s but by reheating: the tabulated $T_{\text{reh}}^*(N)$ grows as $e^{3.05N}$, so $T_{\text{reh}}^*(56) \simeq 7.8 \times 10^{15} \text{ GeV}$ already exceeds the instantaneous-reheating ceiling $T_{\text{reh}}^{\text{inst}} = (30\rho_{\text{end}}/\pi^2 g_*)^{1/4} \simeq 2.5 \times 10^{15} \text{ GeV}$ obtained at $N = 56$; hence $N_{\text{max}} \simeq 55.6$ and $N = 56, 57, 58$ are physically excluded (Appendix D).

Two-sided physical window.

The admissible band is bounded on *both* sides by physics, not by the n_s likelihood alone. On the high side, $T_{\text{reh}}^*(N) \propto e^{3N}$ while the instantaneous-reheating ceiling falls only as $N^{-1/2}$, giving $N_{\text{max}} \simeq 55.6$ as shown above. On the low side, the model cannot be pushed to small N : the tensor-to-scalar ratio falls to $r = 0.01$ at $N_* \simeq 32$ in the locked- N convention (the attractor closed form $r \simeq 8/(\beta^2 N^2)$ gives $N_* \simeq 35$), so $r > 0.01$ requires $N \gtrsim 32$, for which the matching condition (14) returns $T_{\text{reh}} \simeq 2.5 \times 10^{-16} \text{ GeV}$ —more than thirteen orders of magnitude below the BBN floor of

10 MeV—so such a point is physically impossible for any reheating mechanism. The viable interval is therefore $N \in [45, 55.6]$, and Planck 2σ on n_s happens to select almost exactly that range. This is what makes the model falsifiable in a definite way: a LiteBIRD-class detection $r > 0.01$ would demand $N \lesssim 32$ and hence a reheating temperature incompatible with BBN, so it would exclude the model class outright rather than merely disfavour it.

Reheating equation of state and the N – T_{reh} mapping.

The mapping above assumes the condensate behaves as pressureless dust ($w = 0$) throughout the post-inflationary phase. This is an approximation: the potential is not exactly harmonic at the largest amplitudes (Appendix D), and the exact integration of Sec. D 0 gives $\langle w \rangle \simeq -0.02$ averaged over the first few e -folds, approaching zero thereafter. A constant- w reheating epoch with $\bar{w}_{\text{eff}}$ shifts the e -fold count by $\Delta N = \frac{1-3\bar{w}_{\text{eff}}}{12(1+\bar{w}_{\text{eff}})} \ln(T_{\text{reh}}/T_{\text{end}})$; for $\bar{w}_{\text{eff}} \in [0, 1/3]$ this gives $|\Delta N| \lesssim 1$, so the mapping is stable at the stated $\mathcal{O}(1)$ level and the quoted band is unaffected.

Quantum diffusion.

During slow-roll, quantum fluctuations cause the field to undergo a random walk with variance $\langle \delta\chi^2 \rangle \simeq H_{\text{inf}}^3 t / (4\pi^2)$ per Hubble time. Over one e -fold $\Delta t = H_{\text{inf}}^{-1}$, the accumulated quantum jitter is $\Delta\chi_{\text{quant}} \sim H_{\text{inf}}/(2\pi)$, while the classical drift is $\Delta\chi_{\text{class}} \sim \dot{\chi}/H_{\text{inf}} \sim M_{\text{Pl}}\sqrt{2\epsilon}$. The ratio $\Delta\chi_{\text{quant}}/\Delta\chi_{\text{class}} \sim H_{\text{inf}}/(2\pi M_{\text{Pl}}\sqrt{2\epsilon}) \sim 6 \times 10^{-6}/\sqrt{\epsilon}$ is utterly negligible except at the very end of inflation ($\epsilon \rightarrow 1$), where it reaches $\sim 6 \times 10^{-6}$ —still five orders below unity. Stochastic effects therefore do not invalidate the classical slow-roll trajectory for this model.

Attractor forms and non-Gaussianity.

Large-field attractor expressions $n_s \simeq 1 - 2/N$, $r \simeq 8/(\beta^2 N^2)$, $n_t = -r/8$ are useful analytic guides; numerical values quoted in Table 1 use the exact integral (15) and potential slow roll (16) at the locked N . Being of the Starobinsky type, the model also predicts negligible primordial non-Gaussianity: $f_{\text{NL}}^{\text{local}} = -\frac{5}{12}(1 - n_s) \sim -0.02$, with equilateral and orthogonal shapes suppressed by the slow-roll parameters to $\mathcal{O}(10^{-2})$, all well within Planck 2018 constraints [16] ($f_{\text{NL}}^{\text{local}} = -0.9 \pm 5.1$).

3.2 Slow-roll parameters and observables

In the $\chi \gg M_{\text{Pl}}$ limit, the potential slow-roll parameters are

$$\epsilon_V \equiv \frac{M_{\text{Pl}}^2}{2} \left(\frac{V'_E}{V_E} \right)^2 \simeq 2\beta^2 e^{-2\beta\chi/M_{\text{Pl}}} \ll 1, \quad (20)$$

$$\eta_V \equiv M_{\text{Pl}}^2 \frac{V''_E}{V_E} \simeq -2\beta^2 e^{-\beta\chi/M_{\text{Pl}}}, \quad (21)$$

$$N \simeq \frac{e^{\beta\chi_N/M_{\text{Pl}}}}{2\beta^2}. \quad (22)$$

Table 1 Exact potential slow-roll at $\xi = 11.1$ under the locked definition $N = \ln(a_{\text{end}}/a_*)$ (Eqs. (13)–(16)). λ_0 is fixed by $A_s = 2.100 \times 10^{-9}$. T_{reh}^* is the temperature for which cosmological matching (14) returns $N_{\text{derived}} = N$. Planck 2σ allows $N \approx 45$ –56 ($r \leq 0.0052$); with the instantaneous-reheating ceiling the viable interval is $N \in [45, 55.6]$ (Appendix D).

N	λ_0	n_s	r	H_{inf} [GeV]	T_{reh}^* [GeV]
48	7.24×10^{-8}	0.9600	0.00459	1.70×10^{13}	2.5×10^5
49	6.96×10^{-8}	0.9608	0.00441	1.67×10^{13}	5.2×10^6
50	6.70×10^{-8}	0.9616	0.00425	1.64×10^{13}	1.1×10^8
51	6.45×10^{-8}	0.9623	0.00409	1.61×10^{13}	2.2×10^9
52	6.21×10^{-8}	0.9630	0.00394	1.58×10^{13}	4.4×10^{10}
55	5.58×10^{-8}	0.9650	0.00355	1.50×10^{13}	3.8×10^{14}

We use the potential slow-roll (PSR) formulation throughout; the Hubble slow-roll (HSR) parameters $\epsilon_H \equiv -\dot{H}/H^2$ and $\eta_H \equiv \ddot{\chi}/(H\dot{\chi})$ coincide with ϵ_V and η_V to leading order in slow-roll, with relative differences $\Delta\epsilon/\epsilon \sim \mathcal{O}(\epsilon, \eta) \leq 10^{-2}$ in the deep slow-roll regime and rising to $\sim \mathcal{O}(1)$ only at the very end of inflation ($\epsilon \rightarrow 1$). The e -fold number N —the quantity that enters the CMB observables—is insensitive to the PSR/HSR distinction at the ≤ 0.1 level, well within the $\mathcal{O}(1)$ systematic uncertainty already quoted for the post-inflationary phase.

Cutoff disclaimer.

The slow-roll expressions are leading-order results in the Einstein-frame EFT ($\Lambda_E \sim M_{\text{Pl}}$). Higher-derivative corrections enter at order $(H_{\text{inf}}/M_{\text{Pl}})^2 \sim 3 \times 10^{-11}$; the functional forms $n_s(N)$ and $r(N)$ are insensitive to the cutoff.

Perturbation stability.

In the Einstein frame the scalar field χ is canonically normalized with a standard kinetic term $\frac{1}{2}(\partial\chi)^2$, so the scalar sound speed is $c_s^2 = 1$ identically—no gradient instability can arise. The tensor sector is standard GR with $c_T = 1$ (see also Discussion #16). The comoving curvature perturbation ζ is conserved on superhorizon scales for adiabatic single-field models [19]; no entropy/isocurvature modes are sourced because the single field Φ drives both the background and the perturbations. The standard single-field slow-roll formulas for n_s , r , and the scalar power spectrum therefore apply without modification.

The scalar spectral index with NLO correction is

$$n_s \simeq 1 - \frac{2}{N} - \frac{3}{2N^2} + \mathcal{O}(N^{-3}). \quad (23)$$

(The NLO coefficient $-\frac{3}{2}$ applies in the Starobinsky limit $\beta = \sqrt{2/3}$; for general β it acquires a β -dependence. **All tabulated n_s and r in this paper use the exact potential slow-roll evaluation (16) at the locked N , not this NLO closed form.** A quantitative slow-roll scan shows the NLO closed form differs from exact PS by $\leq 3 \times 10^{-4}$ in n_s at $\xi = 11.1$ ($\beta = 0.810$) and throughout $\xi \geq 5$ —well below Planck

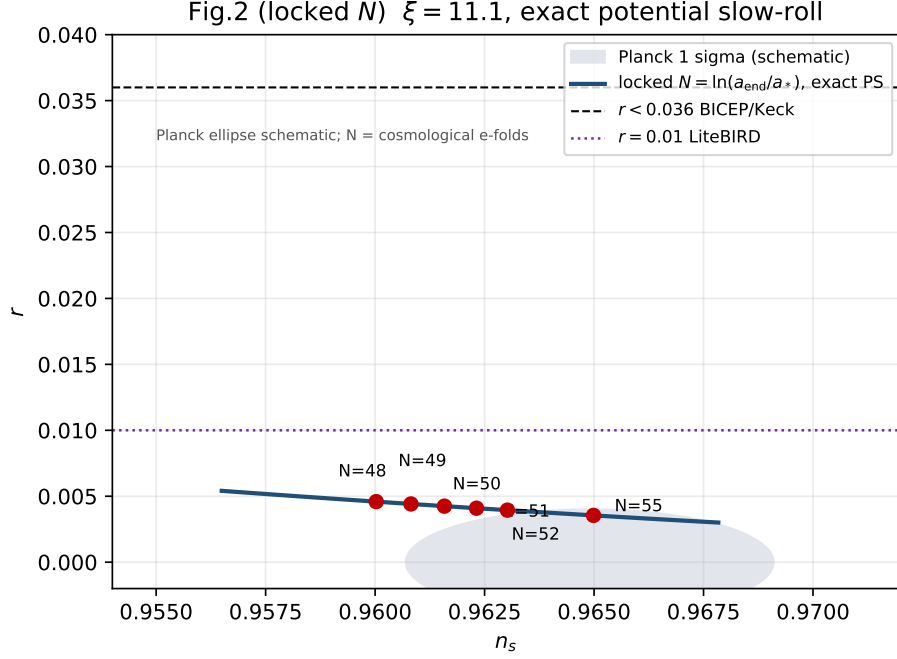


Fig. 2 n_s - r under the locked definition $N = \ln(a_{\text{end}}/a_*)$ at $\xi = 11.1$ (exact potential slow-roll, λ_0 from A_s ; Table 1). Markers: locked- N points. Planck ellipse schematic; BICEP/Keck $r < 0.036$; LiteBIRD line $r = 0.01$. Regenerated by `scripts/fig2_ns_r.py` using the locked- N grid from `lock_n_convention.py`.

$\sigma(n_s) = 0.0042$ —but the shift grows toward $\xi \rightarrow 1$; precision comparisons at small ξ must use the exact integral (15).)

For the locked definition, exact potential slow roll at $\xi = 11.1$ gives $n_s \simeq 0.9616$ at $N = 50$ (-0.79σ from Planck) and $n_s \simeq 0.9650$ at $N = 55$ ($+0.02\sigma$); see Table 1. Planck 2018 [16]: 0.9649 ± 0.0042 (1σ). The prediction approaches the Planck central value as N increases; the 2σ band corresponds to $N \approx 45$ – 56 in this convention.

The tensor-to-scalar ratio under the same convention is $r \simeq 0.00425$ at $N = 50$ and $r \simeq 0.00355$ at $N = 55$; across Planck 2σ $N \approx 45$ – 56 , $r \in [0.0034, 0.0052]$. Planck+BICEP/Keck [16, 17]: $r < 0.036$ (95% CL).

A measurement $r > 0.01$ at 95% CL by LiteBIRD [18] would rule out the entire physically allowed model class in this convention: exact slow-roll inversion puts $r = 0.01$ at $N \approx 32$ (for $\xi = 11.1$), outside the Planck-compatible band, and reheating temperatures required for such small N fall far below BBN. Throughout the 2σ window, $r \leq 0.0052$ at $\xi = 11.1$ (only weakly ξ -dependent once $x_*(N)$ is fixed by the exact integral).

Figure 2 shows the model predictions superimposed on Planck+BICEP/Keck-style constraints.

4 End of slow-roll and reheating

4.1 End of slow-roll and coherent oscillations

Slow-roll ends at $\epsilon = 1$. The exact ϵ derived from $V_E = V_0(1 - e^{-\beta\chi/M_{\text{Pl}}})^2$ is $\epsilon_{\text{exact}} = 2\beta^2 e^{-2x}/(1 - e^{-x})^2$ with $x \equiv \beta\chi/M_{\text{Pl}}$; the large-field limit $(1 - e^{-x}) \approx 1$ reduces it to Eq. (22). Solving $\epsilon_{\text{exact}} = 1$ with $u \equiv e^{-x}$ gives $u = 1/(1 + \sqrt{2}\beta) \approx 0.466$, so

$$e^{-\beta\chi_{\text{end}}/M_{\text{Pl}}} = \frac{1}{1 + \sqrt{2}\beta} \approx 0.466,$$

$$V_{\text{end}} = V_0 \left(1 - e^{-\beta\chi_{\text{end}}/M_{\text{Pl}}}\right)^2 \approx 0.285 V_0 \approx 1.36 \times 10^{63} \text{ GeV}^4 \quad (24)$$

$$(V_0 \simeq 4.78 \times 10^{63} \text{ GeV}^4 \text{ at locked } N = 50).$$

Thereafter the field oscillates around Φ_0 with mass $m_\Phi = \sqrt{2\lambda_0}\Phi_0 \approx 2.7 \times 10^{14} \text{ GeV}$. Near the minimum, $V_J(\Phi) \approx \lambda_0\Phi_0^2(\Phi - \Phi_0)^2 + \mathcal{O}((\Phi - \Phi_0)^3)$, i.e. the potential is quadratic to leading order; the quartic $(\Phi^2 - \Phi_0^2)^2$ form produces a radiation-like equation of state ($w \simeq 1/3$) only for oscillations around $\Phi = 0$, which is never reached since the field is confined to one vacuum branch ($+\Phi_0$ or $-\Phi_0$) by the initial-condition locking of Sec. II. Consequently the amplitude redshifts as $A(t) \propto a^{-3/2}$ and the condensate evolves as pressureless dust ($w = 0$). This last statement is an approximation: the post-inflationary excursion reaches $|x| \equiv |\beta\chi/M_{\text{Pl}}| \simeq 0.88$ (Sec. III and App. D), where the cubic and quartic corrections are $\mathcal{O}(1)$, and the exact integration gives $\langle w \rangle \simeq -0.02$ averaged over the first e -folds rather than exactly zero. The a^{-3} scaling, the N - T_{reh} mapping and the dark-energy budget are therefore accurate at the few-per-cent level, well inside the $\mathcal{O}(1)$ matching uncertainty quoted in Sec. III. We adopt the standard approximation that the transition from slow-roll to coherent oscillations is instantaneous; a more detailed treatment of the transient post-inflationary phase (wherein the field may traverse the steep region of the potential before settling into harmonic motion) would shift the numerical mapping between N and T_{reh} by an $\mathcal{O}(1)$ factor in the expansion history and is deferred to a dedicated lattice study.

Overshoot across the barrier.

At the end of inflation the kinetic energy of the field is $K_{\text{end}} = \frac{1}{2}\dot{\chi}_{\text{end}}^2 = \frac{\epsilon_H}{3-\epsilon_H}V_{\text{end}}$, where $\epsilon_H \equiv -\dot{H}/H^2$; the slow-roll relation $K = \epsilon V$ is only valid to leading order in ϵ , and at $\epsilon_V = 1$ the exact attractor gives $\epsilon_H \simeq 0.499$, hence $K_{\text{end}} \simeq 0.199 V_{\text{end}} \simeq 2.7 \times 10^{62} \text{ GeV}^4$ (obtained by numerically integrating the exact Klein-Gordon equation from the slow-roll regime to $\epsilon_V = 1$; see the discussion of numerical validation in Appendix D). The Jordan-frame barrier height at $\Phi = 0$ is $\Delta V_J = V_J(0) - V_J(\Phi_0) = \lambda_0\Phi_0^4/4 \equiv V_0 \simeq 4.78 \times 10^{63} \text{ GeV}^4$. Although the Jordan-frame barrier at $\Phi = 0$ has height $\Delta V_J = V_0$, that is *not* the quantity governing the dynamics: the propagating degree of freedom is the canonical field χ in the Einstein frame, whose potential is $V_E(x) = V_0(1 - e^{-x})^2$ with $x \equiv \beta\chi/M_{\text{Pl}}$, and the conformal factor pushes the $\Phi \rightarrow 0$ limit to $V_E \rightarrow \infty$. The correct criterion is therefore whether the conserved total energy density suffices to reach $\Phi = 0$, which would require infinite energy. The field is

bounded instead by the turning points of $V_E(x) = \rho_{\text{end}}$, where $\rho_{\text{end}} = K_{\text{end}} + V_{\text{end}} = 1.199 V_{\text{end}} = 0.342 V_0$; solving $(1 - e^{-x})^2 = 0.342$ gives $x_{\text{turn}} = +0.879$ and -0.461 , i.e. $\Phi/\Phi_0 = e^{x/2} \in [0.79, 1.55]$. The field therefore oscillates between $\Phi \simeq 0.79 \Phi_0$ and $\Phi \simeq 1.55 \Phi_0$ and never approaches $\Phi = 0$: no vacuum-branch crossing occurs. (The ratio $K_{\text{end}}/\Delta V_J \simeq 0.057$ is sometimes quoted here, but it is *not* the relevant criterion, since it compares a kinetic energy with a potential difference evaluated in a different frame; the Jordan-frame barrier height ΔV_J is likewise not the dynamical barrier. The barrier is the full height V_0 between the two vacua.) The field displacement from Φ_{end} to $\Phi = 0$ ($\varphi \rightarrow -\infty$) would require traversing the entire plateau, and Hubble friction during the transient post-inflationary phase removes whatever energy remains above the turning point. The initial-condition locking of Sec. II (field mass $m_\Phi \gg H_{\text{inf}}$ during inflation) independently guarantees that the field is exponentially unlikely to be found near $\Phi = 0$ at the end of slow-roll.

4.2 Vanishing of perturbative Yukawa reheating

The same conformal identity $\Omega = \Phi/\Phi_0$ that screens the fifth force (Sec. II E) also governs reheating. The tree-level Einstein-frame $\chi\bar{\psi}\tilde{\psi}$ coupling vanishes exactly (Eq. (10)), so there is no perturbative inflaton decay into fermions via the direct Yukawa channel—this holds for any value of g , including the value $g \simeq 1.5 \times 10^{-7}$ selected by the dark-matter abundance in Sec. V. For $m_\psi \gtrsim m_\chi/2$ (i.e. $g \gtrsim \text{few} \times 10^{-5}$) the decay $\chi \rightarrow \psi\bar{\psi}$ is *additionally* kinematically forbidden, so the closure is then doubly protected; at the lighter abundance-matched value the kinematic condition fails and the closure rests on the vanishing vertex alone. Because Φ oscillates about $\Phi_0 \neq 0$, the Jordan-frame fermion mass $m_\psi = g|\Phi|$ never crosses zero; for $g \geq 10^{-4}$ it varies adiabatically ($\dot{m}_\psi/m_\psi^2 \sim 4.4 \times 10^{-4}/g \ll 1$), while for $g \lesssim 10^{-4}$ —including the dark-matter value $g \simeq 1.5 \times 10^{-7}$ —the variation is non-adiabatic and $\psi\bar{\psi}$ pair production opens. This is the Jordan-frame image of the Einstein-frame gravitational production that sources the dark matter of Sec. V; like gravitational particle production, it is a classical-background/quantum-vacuum effect and does *not* use the vanishing perturbative vertex, so it does not re-open the reheating channel. The scalar self-coupling $\lambda_{\text{self}} \sim \lambda_0 \beta^4/(4\xi^2) \sim 4 \times 10^{-11}$ is too small for efficient self-resonance; a full Floquet analysis of preheating is deferred.

4.3 Gravitational reheating and the status of N

With the perturbative Yukawa and self-resonance channels absent or negligible, reheating proceeds through gravitational mechanisms [21, 23]. We distinguish the relevant channels carefully, because the status of the graviton-pair channel depends on the gravitational sector’s structure:

- (i) **Non-perturbative gravitational particle production** at the end of slow-roll: the rapidly varying metric excites SM fields from the vacuum, providing an initial radiation pulse with $\rho_{\text{rad}}(a_{\text{end}}) \sim 2 \times 10^{-13} \rho_\chi(a_{\text{end}})$. As shown below, this pulse dilutes *faster* than the condensate and therefore cannot reheat the universe on its own; it supplies only a seed bath. The reheating temperature is set by the continuous conformal-anomaly channel (iv).

- (ii) **Perturbative decay of the oscillating condensate to fermion pairs via graviton exchange** ($\chi \rightarrow f\bar{f}$ through an s -channel graviton): this proceeds at $\Gamma_{\chi \rightarrow f\bar{f}}^{(\text{grav exch})} \sim m_\chi^5/M_{\text{Pl}}^4 \sim 10^{-6}$ GeV, slower than channel (i) by $\sim 10^6$ and subdominant for setting T_{reh} . The direct $\chi \rightarrow f\bar{f}$ channel through the residual $\chi f\bar{f}$ coupling $\sim m_f/M_{\text{Pl}}$ (Eq. (42)) is even slower, $\sim 10^{-20}$ GeV for the top quark.
- (iii) **Tree-level decay to graviton pairs** ($\chi \rightarrow h\bar{h}$): this channel is *absent* in the present model. The reason is structural: induced gravity has the $F(\Phi)R/2$ form with $F = \xi\Phi^2$, so after conformal transformation the Einstein-frame Ricci term carries a *constant* coefficient $M_{\text{Pl}}^2/2$ with no χ dependence; the canonical field χ is therefore minimally coupled to the Einstein-frame gravitons (coupling only through the kinetic term), and no $\chi h\bar{h}$ vertex is generated. Consequently $\Gamma(\chi \rightarrow h\bar{h}) = 0$ at tree level. At the 1-loop level, graviton loops induce an effective $\chi h\bar{h}$ vertex of size $\Gamma_{\chi \rightarrow h\bar{h}}^{\text{eff}} \sim m_\chi^5/(16\pi^2 M_{\text{Pl}}^4) \sim 10^{-8}$ GeV, $\sim 10^8$ below channel (i) and $\sim 10^2$ below channel (ii), so this 1-loop contribution—the leading order for this structurally closed channel—is far below the dominant reheating channels and does not affect T_{reh} . This differs from R^2 inflation [4], where the R^2 term produces a scalaron with a χ -dependent $\tilde{R}$ coefficient ($\propto e^{2\chi/\sqrt{6}M_{\text{Pl}}}$), opening the $\chi \rightarrow h\bar{h}$ channel with $\Gamma \sim m_\chi^3/(192\pi M_{\text{Pl}}^2)$; that result does *not* carry over to the $F(\Phi)R/2$ theories (induced gravity or Higgs inflation), for which $\chi \rightarrow h\bar{h}$ is closed.
- (iv) **Continuous conformal-anomaly decay**: although the gauge kinetic term is classically conformally invariant, the trace anomaly $\langle T_\mu^\mu \rangle = (\beta_g/2g)\langle F^2 \rangle$ generates an effective χ -gauge vertex $\mathcal{L}_{\text{eff}} \supset (b_s\alpha_s/4\pi)(\chi/(M_{\text{Pl}}\sqrt{6+1/\xi}))G_{\mu\nu}^a G^{a\mu\nu}$ (and analogously for photons). The corresponding decay rates are $\Gamma_{\chi \rightarrow gg} = b_3^2\alpha_s^2 m_\chi^3/(16\pi^3 M_{\text{Pl}}^2(6+1/\xi))$ and $\Gamma_{\chi \rightarrow \gamma\gamma} \sim \mathcal{O}(10^{-2})$ GeV, giving a total perturbative width $\Gamma_{\text{anom}} \sim \mathcal{O}(10^{-1})$ GeV. Evaluating the QCD channel with the one-loop running coupling, $\alpha_s(m_\chi) = 0.0262$ at $m_\chi = 3.25 \times 10^{13}$ GeV for $b_3 = 7$ (SM value; $\alpha_s(m_Z) = 0.1179$), gives $\Gamma_{\chi \rightarrow gg} \simeq 0.065$ GeV. Unlike channel (i), this is a genuine continuous perturbative decay, so the standard constant-rate reheating formula applies: $T_{\text{reh}}^{(\text{anom})} = (90\Gamma_{\text{anom}}^2 M_{\text{Pl}}^2/(\pi^2 g_*))^{1/4} \simeq 2.1 \times 10^8$ GeV. Throughout this work we retain $T_{\text{reh}} = 10^9$ GeV as the fiducial reference temperature (it is the value entering Table 1 and the N mapping below); with the physical running coupling the anomaly channel gives 2.1×10^8 GeV instead, which shifts the corresponding e -fold number by only $\Delta N \simeq 0.5$ (i.e. $N \simeq 50.2$ rather than 50.7), well inside the $\mathcal{O}(1)$ matching uncertainty declared in Sec. III B. We therefore continue to quote $N \simeq 50.7$ as the primary reference point.

The parametric estimate below accordingly combines channel (iv); channels (ii) and (iii) do not alter the reheating temperature at leading order. A precise prediction of T_{reh} and N cannot be obtained from a perturbative decay rate alone: the Einstein-frame $\chi\psi\bar{\psi}$ vertex vanishes exactly, the $\chi \rightarrow h\bar{h}$ channel is structurally closed, and the Jordan-frame Yukawa channel is kinematically forbidden. With the perturbative channels closed, N is not microphysically predicted but is constrained by the consistency requirements that the universe reheats to a temperature compatible with BBN ($T_{\text{reh}} \geq 10$ MeV) and does not overproduce relics. The window

is bounded from below by the BBN floor and from above by instantaneous reheating, $T_{\text{reh,max}} = (30\rho_{\text{end}}/(\pi^2 g_*))^{1/4} \simeq 2.6 \times 10^{15} \text{ GeV}$ at $N = 50$ (equivalently $T_{\text{reh,max}} \simeq 0.430 V_{\text{end}}^{1/4}$, where $V_{\text{end}}^{1/4} \simeq 6.1 \times 10^{15} \text{ GeV}$); this ceiling is what limits the n_s -allowed band to $N \leq 55.6$ (Appendix D).

Using the cosmological definition (13) and matching (14) (rather than a single closed-form $N(T_{\text{reh}})$), exact slow-roll inversion with A_s normalization gives the T_{reh}^* column of Table 1. Planck 2σ on n_s then selects

$$N \approx 45\text{--}56 \quad (2\sigma \text{ Planck}), \quad \text{anomaly } T_{\text{reh}} \sim 10^9 \text{ GeV} \Rightarrow N \simeq 50.7, \quad (25)$$

with $r \leq 0.0052$ throughout that band at $\xi = 11.1$. The self-consistent point for anomaly-dominated reheating is therefore $N \simeq 50.7$ in this convention, not an a-priori fixed $N = 50$. Matching assumptions (pivot k_* , equation of state during reheating, g_*) move N by $\mathcal{O}(1)$; we quote the locked-definition numbers as the primary results and treat attractor closed forms as approximations only.

A single closed-form matching estimate, $N = 61 + \frac{1}{4} \ln(V_{\text{end}}/M_{\text{Pl}}^4) - \frac{1}{3} \ln(V_{\text{end}}^{1/4}/T_{\text{reh}})$ (Liddle–Leach style [20]), remains a useful cross-check, but the physical window used here follows from the locked definition (13) and Planck n_s , Eq. (25).

All N -dependent numerical values in this work refer to the locked definition (13). Primary quoted points are the anomaly-reheating self-consistency value $N \simeq 50.7$ and the illustrative $N = 50$ row of Table 1; attractor closed forms $n_s \simeq 1 - 2/N$, $r \simeq 8/(\beta^2 N^2)$ are analytic guides only and are not mixed with the table.

a. Gravitational production and conformal-anomaly decay.

We now identify which channels can reheat the universe. After inflation, the oscillating χ condensate dominates the energy density. Two distinct radiation sources operate:

(a) *Initial gravitational pulse* (channel (i)): Standard results [22, 23] give an initial radiation density $\rho_{\text{rad}}(a_{\text{end}}) \sim N_{\text{eff}} H_{\text{inf}}^4 / (192\pi^2)$, where $N_{\text{eff}} \sim \mathcal{O}(10)$ counts SM degrees of freedom with non-conformal coupling. With $H_{\text{inf}} \simeq 1.64 \times 10^{13} \text{ GeV}$ (locked $N = 50$; $1.61 \times 10^{13} \text{ GeV}$ at $N = 51$), this yields $\rho_{\text{rad}}(a_{\text{end}}) \sim 4 \times 10^{50} \text{ GeV}^4$, compared to the exact condensate density $\rho_{\text{end}} = K_{\text{end}} + V_{\text{end}} = 1.199 V_{\text{end}} \simeq 1.64 \times 10^{63} \text{ GeV}^4$ —a fractional radiation energy density of only $\sim 2.3 \times 10^{-13}$. This pulse does *not* by itself reheat the universe. During condensate domination the condensate redshifts as a^{-3} while the pulse redshifts as a^{-4} , so the radiation *fraction* decreases:

$$\frac{\rho_{\text{rad}}}{\rho_{\chi}} = \frac{R/a^4}{Y/a^3} = \frac{R}{Y} \frac{1}{a} \propto a^{-1}, \quad (26)$$

with $Y \equiv \rho_{\chi} a^3$ and $R \equiv \rho_{\text{rad}} a^4$ both conserved in the absence of decay. The two densities therefore never cross; if the condensate did not decay, the universe would remain matter-dominated, with a radiation temperature today of only $\sim 4 \times 10^{-10} \text{ eV}$. The pulse is thus a *seed* bath, not a reheating channel, and a genuine decay of the condensate (channel (iv)) is mandatory.

(b) *Continuous conformal-anomaly decay* (channel (iv)): the rates and $T_{\text{reh}}^{(\text{anom})} \simeq 2.1 \times 10^8 \text{ GeV}$ are as above.

Combined picture.

Reheating is driven *solely* by channel (iv); channel (i) provides only a seed radiation bath that dilutes away relative to the condensate (Eq. (26)) and does not by itself produce a radiation-dominated epoch. The two are nevertheless physically distinct—channel (i) is a one-time non-perturbative burst driven by the rapid metric change at the end of slow-roll, while channel (iv) is a continuous perturbative decay during the subsequent oscillatory phase—so no double-counting arises. Because the seed bath is negligible compared with the decay products, T_{reh} is set by channel (iv) alone to within $\lesssim 10^{-3}$. The fiducial value $T_{\text{reh}} \simeq 10^9$ GeV (or 2.1×10^8 GeV with the physical running coupling) is set by the continuous anomaly decay; under the locked N definition this corresponds to $N \simeq 50.7$ (Table 1).

Validity of the homogeneous-condensate approximation.

The reheating estimates above treat the oscillating χ field as a spatially homogeneous condensate. In a full preheating scenario, quantum fluctuations and non-linear resonance effects can fragment the condensate into inhomogeneous modes; when fragmentation is efficient, the uniform approximation breaks down and the energy transfer to radiation can be significantly faster than the perturbative estimate. For the present model, fragmentation is suppressed by two features: (i) the self-resonance parameter $\lambda_{\text{self}} \sim 4 \times 10^{-11}$ (Sec. IV) is too small to drive efficient χ fragmentation; (ii) the coupling to SM decay products is $\leq 10^{-17}$, so backreaction from produced radiation on the condensate is negligible. The homogeneous approximation is therefore well justified for the radiation-producing channel (iv). For the dark-matter fermion ψ , whose coupling $g \simeq 1.5 \times 10^{-7}$ is $\sim 10^{10}$ times larger than $g_{\chi ff}$, the homogeneous treatment is sufficient for the absolute normalization: the mode-equation integration now extends through the oscillating-condensate era on the exact homogeneous background (Appendix D0), and the two effects a lattice would capture—fragmentation ($\lambda_{\text{self}} \sim 4 \times 10^{-11}$, too weak to matter) and backreaction ($\rho_\psi/\rho_{\text{end}} \sim 10^{-19}$)—are parametrically negligible (Discussion #3). Non-perturbative resonance effects could raise T_{reh} by one to two orders of magnitude, but a precise calculation requires a full lattice simulation. The perturbative fermion channel (channel (ii), $\Gamma \sim 10^{-6}$ GeV) is subdominant; the graviton-pair channel (iii) is structurally closed; the open anomalous gauge channel is a robust feature of any scalar-tensor theory with $F(\Phi)R/2$ coupling and SM embedding, analogous to the scalaron decay in R^2 inflation [4].

Parametric resonance into SM fermions and gauge bosons.

During the oscillatory phase, the time-varying conformal factor $\Omega(t) = \Phi(t)/\Phi_0$ induces a periodic modulation of SM particle masses: $m_f(t) = m_f^{(0)}/\Omega(t)$ for Higgs-Yukawa fermions and $\tilde{m}_W(t) = m_W/\Omega(t)$ for massive gauge bosons (Sec. VII). In principle, such modulated mass terms can drive non-perturbative parametric resonance (Floquet/Mathieu instability), potentially competing with the anomalous decay channel (iv). However, the oscillation amplitude at the bottom of the potential is $\delta\Phi/\Phi_0 \sim 10^{-111}$ (Sec. VI), so the fractional modulation $\Delta m_f/m_f \sim 10^{-111}$ is utterly negligible after reheating. During the large-amplitude oscillation phase ($\chi \sim M_{\text{Pl}}$, immediately after the end of slow-roll), the modulation is $\mathcal{O}(1)$, but

parametric resonance is driven by the *mass modulation* $\delta m_f/m_f \sim \delta\Phi/\Phi_0 \sim \mathcal{O}(1)$ rather than by the Yukawa coupling $g_{\chi ff}$; the correct Mathieu parameter [21] is $q \sim (\delta\Phi/\Phi_0)^2(m_f/m_\chi)^2 \sim (m_f/m_\chi)^2$. For the top quark ($m_t = 173 \text{ GeV}$, $m_\chi \simeq 3.25 \times 10^{13} \text{ GeV}$) this gives $q \sim 2.8 \times 10^{-23}$, and for lighter species it is exponentially smaller—over twenty orders below the broad-resonance threshold $q \geq 0.1$. The gauge-boson channel is likewise suppressed by $\sim (m_W/M_{\text{Pl}})^2 \leq 10^{-33}$. SM fermion and gauge-boson parametric resonance is therefore completely negligible for reheating; the dominant radiation sources remain the gravitational pulse (i) and the conformal-anomaly decay (iv). The absolute normalization of the dark-matter relic abundance is obtained from the homogeneous mode-equation computation through the oscillating-condensate era (Sec. V, Appendix D0); a Floquet/lattice simulation would serve as the 3D cross-check of the homogeneous assumption (Discussion #3).

4.4 Post-reheating cosmology

After reheating, the universe cools and follows standard Λ CDM. The thermalization rate $\Gamma_{\text{therm}}/H \sim 10^6$ at $T \sim 10^9 \text{ GeV}$ (see Discussion #17) ensures that the radiation bath reaches kinetic and chemical equilibrium within a few Hubble times. Non-thermal relics from gravitational particle production—including any high-energy tails of the fermion and gauge-boson spectra—are diluted by the dominant thermal bath; their residual non-thermal contributions to BBN light-element abundances are suppressed by the energy-density ratio $\rho_{\text{non-th}}/\rho_{\text{rad}} \leq 10^{-13}$ and are far below current observational precision. The universe enters BBN with the standard radiation-dominated initial conditions, and the subsequent thermal history is indistinguishable from Λ CDM.

5 Fermionic dark matter from gravitational production

5.1 Setup: the Yukawa fermion ψ

The Yukawa fermion ψ already present in the action ($\mathcal{L}_Y = -g\Phi\bar{\psi}\psi$, Sec. II) is promoted to the dark matter. We take ψ to be a *Dirac, gauge-singlet* fermion with no Higgs portal ($\bar{L}\tilde{H}\psi$ set to zero); both conditions are technically natural [11], since setting them to zero increases the symmetry. Its Einstein-frame mass is exact and χ -independent:

$$m_\psi = g\Phi_0 = \frac{gM_{\text{Pl}}}{\sqrt{\xi}}, \quad (27)$$

a consequence of the conformal identity $\Omega = \Phi/\Phi_0$ (Eq. (10)): the Yukawa term transforms as $-g\Phi\bar{\psi}\psi\sqrt{-g} \rightarrow -g\Phi_0\bar{\tilde{\psi}}\tilde{\psi}\sqrt{-\tilde{g}}$ with $\Omega^{1+3-4} = 1$ exactly, so the tree-level $\chi\tilde{\psi}\tilde{\psi}$ vertex vanishes identically for all Φ and all g (Sec. II E).

5.2 Stability of ψ

The fermion ψ is stable under a discrete $\mathbb{Z}_2^\psi$ symmetry under which $\psi \rightarrow -\psi$ (all SM fields and Φ are even). This discrete symmetry may be viewed as the low-energy remnant of a UV $U(1)_D$ gauge symmetry spontaneously broken to $\mathbb{Z}_2^\psi$, and *if* realized

as a gauge symmetry it is respected by all quantum-gravitational effects [24, 25]—in contrast to a global $U(1)$, which quantum gravity is expected to break. The $\mathbb{Z}_2^\psi$ protection operates independently of the conformal decoupling:

- **Gauge-protected stability (conditional).** Under $\mathbb{Z}_2^\psi$, any operator connecting a single ψ to purely even SM final states ($\psi \rightarrow \text{SM}$) is odd and strictly forbidden at all orders. Operators with an even number of ψ fields (e.g. $\psi\bar{\psi}$ pair annihilation) are allowed but irrelevant at late times due to the low ψ abundance. The Dirac mass term $m_\psi\bar{\psi}\psi$ is even and permitted, while single- ψ decay is absolutely forbidden. No additional gauge fields or scalar degrees of freedom are required in the low-energy EFT; the discrete gauge nature of the symmetry is declared at the EFT level, with the UV $U(1)_D \rightarrow \mathbb{Z}_2^\psi$ completion left implicit. For a discrete gauge symmetry to remain exact at the quantum level, the mixed $\mathbb{Z}_2^\psi$ -gravitational and $\mathbb{Z}_2^\psi$ -gauge anomalies must cancel [26, 27]; we assume this condition is satisfied by the UV completion (e.g. through the addition of anomaly-canceling heavy fermions at the $U(1)_D$ breaking scale), which is a standard requirement for any discrete gauge symmetry [24] and does not affect the low-energy cosmology discussed here.
- **No renormalizable decay channel.** Being a gauge singlet, ψ has no renormalizable coupling to SM fields; with the $\chi\psi\psi$ vertex identically zero, the only residual interaction is gravitational (which respects $\mathbb{Z}_2^\psi$). One precision: a Majorana mass term $\frac{1}{2}m_M\psi^c\psi$ is even under $\mathbb{Z}_2^\psi$ ($\psi^c\psi \rightarrow +\psi^c\psi$) and is therefore *not* excluded by it— $\mathbb{Z}_2^\psi$ forbids single- ψ operators (decays), not ψ -pair operators. What keeps ψ Dirac rather than Majorana is the UV charge assignment: we treat the absence of an induced Majorana term as part of the anomaly-free UV-completion assumption declared above.

These two protection mechanisms operate at distinct levels and are not interchangeable: $\mathbb{Z}_2^\psi$ forbids *any* single- ψ decay operator—including those induced by quantum gravity (dimension-5 $\psi\sigma^{\mu\nu}F_{\mu\nu}$, four-fermion operators, etc.)—at all orders, provided the mixed anomalies cancel. The vanishing $\chi\psi\psi$ vertex, by contrast, is a tree-level kinematic feature of the conformal identity; it closes the perturbative reheating channel $\chi \rightarrow \psi\bar{\psi}$ but would not *prevent* ψ decay through, e.g., a graviton-mediated $\psi \rightarrow \nu\gamma$ loop if $\mathbb{Z}_2^\psi$ were absent. Conversely, even if the $\chi\psi\bar{\psi}$ vertex were non-zero (as in a generic scalar-tensor theory), $\mathbb{Z}_2^\psi$ would still guarantee ψ stability, though perturbative reheating into ψ would then be open. The two mechanisms are therefore complementary: $\mathbb{Z}_2^\psi$ provides the irreducible stability guarantee, while the conformally vanishing vertex ensures that reheating does not overproduce ψ through the perturbative channel, leaving gravitational production as the sole source (Sec. IV). This is the structural reason ψ can be dark matter where the condensate χ cannot: χ is a real scalar with no conserved number and a residual SM coupling $g_{\chi ff} \sim m_f/M_{\text{Pl}}$ (Eq. (42)), so it decays in $\tau_\chi \sim 10^{-27}$ s (graviton-mediated) and cannot survive to the present.

Indirect-detection prospects.

Because $\mathbb{Z}_2^\psi$ forbids ψ decay at all orders, there is no indirect-detection signal (gamma rays, neutrinos) from ψ annihilation or decay in the present universe. If quantum-gravity effects at the Planck scale were to induce $\mathbb{Z}_2^\psi$ -breaking operators, the ψ decay lifetime would be $\tau_\psi \sim M_{\text{Pl}}^2/m_\psi^3 \sim 3 \times 10^{-21} \text{ s}$ (for $m_\psi \simeq 1.1 \times 10^{11} \text{ GeV}$), i.e. *instantaneous* on cosmological timescales. Therefore, $\mathbb{Z}_2^\psi$ must be an exact, anomaly-free discrete gauge symmetry—any violation, even Planck-suppressed, would eliminate the dark-matter candidate entirely. This makes the UV completion (see discussion of anomaly-canceling heavy fermions in Sec. V) an essential, not optional, ingredient of the model. Under the assumption that such a UV completion exists and that $\mathbb{Z}_2^\psi$ remains exact at all scales, ψ is stable with no indirect-detection signal.

Direct-detection prospects.

The ψ -nucleon scattering cross section is dominated by graviton t -channel exchange, giving $\sigma_{\psi N} \sim G_N^2 m_N^2 (\hbar c)^2 \sim 10^{-104} \text{ cm}^2$ (using $G_N = 1/(8\pi M_{\text{Pl}}^2)$, $m_N \sim 1 \text{ GeV}$)—far below the XENONnT/LZ sensitivity floor ($\sim 10^{-46} \text{ cm}^2$). Higgs-exchange contributions (through radiatively generated $\lambda_{\Phi H}$) are also negligible: the effective Yukawa $y_\psi^{\text{eff}} \sim g_{\text{loop}} m_\psi / M_{\text{Pl}} \leq 10^{-29}$ yields $\sigma_{\text{SI}} \leq 10^{-80} \text{ cm}^2$. Direct detection is therefore irrelevant for this model. Direct-detection and accelerator searches are likewise insensitive to a fermion of mass $\simeq 1.1 \times 10^{11} \text{ GeV}$ coupled only gravitationally. The observational status of ψ is therefore that of a gravitationally produced superheavy relic whose absolute stability is conditional on the anomaly-free UV completion discussed above, testable only through its relic abundance (Sec. V) and its gravitational imprint on structure formation.

5.3 Closure of the perturbative reheating channel

The same vanishing vertex closes the perturbative $\chi \rightarrow \psi\bar{\psi}$ channel for *any* g , independent of kinematics. On the heavy branch $m_\psi \gtrsim m_\chi/2$ the decay is *additionally* kinematically forbidden, since

$$m_\chi \simeq 3.25 \times 10^{13} \text{ GeV} \approx 2H_{\text{inf}}, \quad 2m_\psi \geq 2H_{\text{inf}} \sim m_\chi, \quad (28)$$

so $m_\chi \leq 2m_\psi$: on that branch the closure is doubly protected. Vertex closure, by contrast, holds for *any* g . Kinematic closure additionally requires $m_\psi \geq m_\chi/2$, i.e. $g \gtrsim \text{few} \times 10^{-5}$ at $\xi = 11.1$; the lighter branch selected by the abundance below ($g \simeq 1.5 \times 10^{-7}$, $m_\psi \simeq 6.6 \times 10^{-3} H_{\text{inf}}$, Sec. V) does not satisfy the kinematic condition and is closed by the vanishing vertex alone. The reheating channel of Sec. IV is therefore never reopened on either branch.

5.4 Production: gravitational particle production

The fermion ψ is produced at the end of inflation by a single physical mechanism—the non-adiabatic change of the spacetime metric at the end of slow-roll—which excites every minimally-coupled field from its vacuum (the same mechanism that produces

the SM radiation of Sec. IV). Gravitational production of superheavy particles in the wimpzilla regime has been studied extensively [28–30], and the present model falls into this class. In the Einstein frame ψ is a free Dirac field of constant mass $m_\psi = g\Phi_0$ with no χ coupling (Sec. V), so its production is governed entirely by the time-dependent Friedmann background; for $m_\psi \leq H_{\text{inf}} \simeq 1.64 \times 10^{13}$ GeV the production is near its peak (wimpzilla regime [22, 23]), with number density

$$n_\psi(a_{\text{end}}) \sim H_{\text{inf}}^3 e^{-2\pi m_\psi/H_{\text{inf}}} \quad (29)$$

up to $\mathcal{O}(1)$ factors, where $n_\psi^{\text{tot}} \equiv n_\psi + n_{\bar{\psi}}$ denotes the total number density of particles plus antiparticles (gravitational production is symmetric under $\psi \leftrightarrow \bar{\psi}$). The exponent follows from the exact Bogoliubov coefficient of a Dirac field in de Sitter, $|\beta_k|^2 = 1/(e^{2\pi m_\psi/H_{\text{inf}}} + 1)$ with index $\nu = \frac{1}{2} \mp i m_\psi/H_{\text{inf}}$ [31, 32]; the factor of 2 relative to a single Boltzmann factor arises because the exact coefficient of a single mode contains two suppressing factors: $|\Gamma(\frac{1}{2} + i\mu)|^2 = \pi/\cosh(\pi\mu) \simeq 2\pi e^{-\pi\mu}$ and the Bunch–Davies normalization of the mode function, $e^{-\pi m_\psi/(2H_{\text{inf}})}$, which enters squared in $|\beta_k|^2$. This process does *not* use the vanishing perturbative $\chi\psi\bar{\psi}$ vertex.

Scope of the estimate.

Equation (29) is the de Sitter-era result and should be regarded as an upper bound on the suppression rather than as the final answer. Two effects go beyond it. First, for a fermion the massless limit is conformal, so gravitational production is suppressed by m_ψ : the production rate *rises* with m_ψ up to $m_\psi \sim m_\chi$ rather than peaking at H_{inf} , and the reheating-era contribution from the oscillating scale factor $a(t)$ is comparable to (and for $m_\psi \gtrsim H_{\text{inf}}$ dominates over) the slow-roll contribution, without the exponential factor of Eq. (29) [31, 33]. Second, in the present model the condensate mass satisfies $m_\chi/H_{\text{inf}} \simeq 1.99$, i.e. the post-inflationary phase is not a fast-oscillation regime ($2\pi H_{\text{inf}}/m_\chi \simeq 3.2$ e-folds per oscillation), so the usual parametric-resonance treatment does not apply either. The abundance quoted below therefore carries an $\mathcal{O}(1)$ (possibly larger) normalization uncertainty; that uncertainty is now *measured* by the exact homogeneous-background computation of Appendix D0 (extended through the condensate era, Sec. V), and the residual is the transition-shape systematic quantified there. In what follows we go one step beyond Eq. (29): the abundance matching is performed with the numerically measured spectrum $n_\psi(m_\psi)$ obtained by integrating the mode equation across the transition (Appendix D0), which for $m_\psi \gtrsim H_{\text{inf}}$ is only weakly dependent on m_ψ rather than exponentially suppressed, and therefore admits a light solution as well.

Jordan-frame cross-check.

The same production admits an equivalent Jordan-frame description in which the metric is rescaled by the time-varying $\Omega = \Phi/\Phi_0$ and the fermion mass $m_\psi(t) = g|\Phi(t)|$ is time-dependent. The adiabaticity parameter at the post-inflationary oscillation

amplitude $\Phi_{\text{osc}} \sim 0.8 \Phi_0$ is

$$\frac{\dot{m}_\psi}{m_\psi^2} \sim \frac{m_\Phi \Phi_{\text{osc}}}{g \Phi_0^2} \sim \frac{4.4 \times 10^{-4}}{g}, \quad (30)$$

so adiabaticity fails for $g \lesssim 10^{-4}$, and at the abundance-matched value $g \simeq 1.5 \times 10^{-7}$ the parameter (30) reaches $\sim 3 \times 10^3$, i.e. strongly non-adiabatic $\psi\bar{\psi}$ pair production—consistent with the light branch found below, which sits at $m_\psi \ll H_{\text{inf}}$. This Jordan-frame picture is the *conformal image* of the Einstein-frame gravitational production above, not an independent channel: both are driven by the same single $\Phi(t)$ dynamics through the conformal relation $\tilde{g}_{\mu\nu} = \Omega^2(\Phi)g_{\mu\nu}$, so they should not be added. The Jordan-frame adiabaticity check serves only as a consistency cross-check on the location of the production peak; the abundance estimate below uses the Einstein-frame result (29) alone. The precise m_ψ that yields the observed abundance is not computable from this parametric estimate alone; it is obtained below from the mode-equation integration, with the oscillating-condensate era included in the exact homogeneous-background computation (Sec. V) and the remaining non-homogeneous effects parametrically negligible ($\lambda_{\text{self}} \sim 4 \times 10^{-11}$, $\rho_\psi/\rho_{\text{end}} \sim 10^{-19}$), so that a lattice simulation is an independent cross-check rather than the missing ingredient. The same parameter also locates the weakest point of that calculation: at $g \simeq 1.5 \times 10^{-7}$ the quantity (30) is $\sim 3 \times 10^3$, some 470 times larger than at the heavy-branch point (32), so the light branch lies deepest in the regime where the post-inflationary mass variation is non-adiabatic and the oscillating condensate—absent from the transition-only calculation—can contribute. A lattice simulation is therefore no longer the only test of the light branch—the controlled-background validation battery of Appendix D0 (convergence, unitarity, massless/conformal limit, and the exact de Sitter initial-data anchor) now supports it—but it remains the only fully non-perturbative one.

5.5 Abundance: mechanism and window

After production ψ is non-interacting and redshifts as matter, $n_\psi \propto a^{-3}$, decoupled from both the decaying condensate and the radiation bath. Diluting $n_\psi(a_{\text{end}}) \sim H_{\text{inf}}^3$ by the entropy dilution factor $(a_{\text{end}}/a_0)^3$ gives, parametrically,

$$\Omega_\psi \sim \frac{m_\psi H_{\text{inf}}^3 e^{-2\pi m_\psi/H_{\text{inf}}} (a_{\text{end}}/a_0)^3}{\rho_c}. \quad (31)$$

The dilution factor scales as $(a_{\text{end}}/a_0)^3 \propto T_{\text{reh}}$ (from $a_{\text{end}}/a_{\text{reh}} \propto T_{\text{reh}}^{4/3}$ during matter-like condensate domination and $a_{\text{reh}}/a_0 \propto T_{\text{reh}}^{-1}$ during radiation domination), so for the fiducial $T_{\text{reh}} \sim 10^9 \text{ GeV}$ we have $(a_{\text{end}}/a_0)^3 \simeq 1.02 \times 10^{-92}$. The exponential suppression lets Ω_ψ be tuned through $\mathcal{O}(1)$ near $m_\psi \sim H_{\text{inf}}$; the precise matching point within

$$m_\psi \simeq 3.1 H_{\text{inf}} \simeq 5.1 \times 10^{13} \text{ GeV}, \quad g = \frac{m_\psi}{\Phi_0} \simeq 6.9 \times 10^{-5}, \quad (32)$$

gives $\Omega_\psi = \Omega_{\text{DM}} \simeq 0.265$ at $T_{\text{reh}} = 10^9$ GeV. The Yukawa coupling g is a free parameter of the model, fixed a posteriori by the dark-matter relic abundance; it is not predicted from first principles, in the same sense that ξ and λ_0 are fitted to A_s and the r bound. Because $(a_{\text{end}}/a_0)^3 \propto T_{\text{reh}}$, a one-order-of-magnitude shift in T_{reh} moves the matching g by only 1.13 (for $u \equiv 2\pi m_\psi/H_{\text{inf}} \simeq 19$, $d \ln g = d \ln T/(u-1)$)—a weak sensitivity specific to this exponential branch, to be contrasted with the light branch found below, where $m_\psi n_\psi \propto m_\psi^2$ gives instead $g \propto T_{\text{reh}}^{-1/2}$, a factor $\sqrt{10}$; across the BBN-allowed $T_{\text{reh}} \in [10^{-2}, 2.6 \times 10^{15}]$ GeV, the matching g varies by a factor of a few, so the gravitational/anomaly reheating estimate $T_{\text{reh}} \sim 10^9$ GeV is what selects $g \simeq 7 \times 10^{-5}$, not a narrow prediction. This matching assumes that the de Sitter exponential (29) dominates the production. As noted above, the reheating-era contribution is *not* exponentially suppressed, so $g \simeq 6.9 \times 10^{-5}$ should be read as the matching point *within that framework* rather than as a prediction; the full mode-equation calculation is deferred (Discussion). Note also that Ω_ψ as a function of g has a maximum at $g = (2\pi)^{-1} H_{\text{inf}}/\Phi_0 \simeq 3.6 \times 10^{-6}$; the matching point is on the large- g branch, and the lower part of the interval $g \sim 10^{-5}$ – 10^{-4} overproduces ψ within this framework; the mode-equation result of the next paragraph supersedes this picture.

One light branch, not two competing branches.

Equation (31) is non-monotonic in g , so the condition $\Omega_\psi = \Omega_{\text{DM}}$ does not by itself single out Eq. (32). Because a massless Dirac field is conformally invariant, gravitational production switches off as $m_\psi \rightarrow 0$; hence $\Omega_\psi(g) \rightarrow 0$ on the light side and a second solution must exist below the maximum at $g = (2\pi)^{-1} H_{\text{inf}}/\Phi_0 \simeq 3.6 \times 10^{-6}$. Replacing the de Sitter exponential (29) by the actual mode-equation spectrum across the end-of-inflation transition (Appendix D0), and fixing the model’s own anomaly-channel temperature $T_{\text{reh}} = 2.1 \times 10^8$ GeV, we find that $\Omega_\psi = \Omega_{\text{DM}}$ is met at $g \simeq 1.0 \times 10^{-7}$ in that transition-only calculation, i.e. $m_\psi \simeq 4.6 \times 10^{-3} H_{\text{inf}} \simeq 7.5 \times 10^{10}$ GeV, where $n_\psi/H_{\text{inf}}^3 \simeq 1.4 \times 10^{-5}$ and $m_\psi n_\psi \propto m_\psi^2$. Over the same spectrum the heavy side is *not* a competing solution: Ω_ψ rises monotonically with m_ψ for $m_\psi \gtrsim H_{\text{inf}}$ (we measure $\Omega_\psi \approx 1.6 \times 10^3$ – 3.0×10^3 over $m_\psi/H_{\text{inf}} = 3$ – 12), so Eq. (32) overproduces ψ by $\sim 6 \times 10^3$ and every heavier value overproduces at least as much. The heavy side becomes viable only if T_{reh} is lowered by some four orders of magnitude—we find $T_{\text{reh}} \simeq 2$ – 3×10^4 GeV over $m_\psi/H_{\text{inf}} \in [1, 12]$ —which the conformal-anomaly channel of Sec. IV does not provide. Within the transition calculation the abundance matching is therefore unique. The direction in which the omitted physics moves g is now measured rather than assumed: extending the mode integration through the oscillating-condensate era on the exact homogeneous background (Appendix D0) raises the production by $R \equiv n_{\text{exact}}/n_{\text{avg}} \simeq 1.24$ at fixed transition, which alone would lower the coupling by $R^{-1/2} \simeq 0.90$, but replacing the instantaneous-splice transition by the real slow-roll descent lowers the transition-only production baseline by a factor $\simeq 2.7$ and dominates, so the net shift is *upward*: $g \simeq 1.5 \times 10^{-7}$ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV, $n_\psi/H_{\text{inf}}^3 \simeq 9.5 \times 10^{-6}$). The transition-only value $g \simeq 1.0 \times 10^{-7}$ is thus a *lower*, not an upper, reference point of the final coupling. Equation (32) is retained as the reference point of the exponential framework rather than as a prediction.

The mechanism is viable and matches the observed abundance on the light branch at $g \simeq 1.5 \times 10^{-7}$ in the exact homogeneous-background computation (transition + oscillating-condensate era, validated by the convergence/unitarity/conformal battery of Appendix D0). In that treatment the split of the condensate energy between radiation and the stable ψ component requires no additional input: ψ is a test field with $\rho_\psi/\rho_{\text{end}} \sim 10^{-19}$ at the matching point, so the two channels are independent; a 3D Floquet/lattice simulation remains as the non-perturbative cross-check (Discussion).

Robustness window, and what remains beyond the homogeneous computation.

The matching point is a two-parameter statement rather than a number. On the light branch $m_\psi n_\psi \propto m_\psi^2$, hence $g \propto T_{\text{reh}}^{-1/2}$: a shift in the reheating temperature moves g and m_ψ but leaves the branch itself intact. Table 2 propagates a factor- 10^2 band in T_{reh} —spanning the physical- α_s anomaly value, the fiducial 10^9 GeV reference, and the possible one-to-two-order non-perturbative enhancement—through the locked- N matching of Sec. III. Across that band N moves by 1.5, n_s by 1.1×10^{-3} (a few per cent of the Planck uncertainty) and r by 5.5%, so the inflationary predictions are *not* hostage to the reheating history. The dark-matter matching point, by contrast, moves by a full factor of 10 in g and m_ψ (through $m_\psi = g\Phi_0$), while the free-streaming length remains 17–18 orders of magnitude below the warm-dark-matter bound throughout: the light branch stays cold dark matter over the whole band. Note that λ_{fs} scales approximately as $m_\psi^{-1/2}$ across the table rather than as m_ψ^{-1} (measured exponent -0.54 to -0.55): the length is accumulated while the particle is relativistic during matter-like condensate domination, so it scales as $\sqrt{a_{\text{NR}}}$ with $a_{\text{NR}} \propto 1/m_\psi$, and the full three-era expansion history—not an instantaneous non-relativistic estimate—enters the integral. The oscillating-condensate contribution, now measured on the exact homogeneous background, indeed raises the production ($R = n_{\text{exact}}/n_{\text{avg}} \simeq 1.24$ at the matching point) and lowers the matching g by $R^{-1/2} \simeq 0.90$ —the branch shifts, it does not fail. But the net shift is set by a different systematic: replacing the instantaneous-splice transition by the real slow-roll descent lowers the transition-only production baseline by a factor $\simeq 2.7$, which dominates and moves the matched coupling to $g \simeq 1.5 \times 10^{-7}$ (Sec. V). Quoting the transition-only baseline $g \simeq 1.0 \times 10^{-7}$ is an abbreviation for a window: over the range $T_{\text{reh}} = 10^8$ – 10^9 GeV spanned by the anomaly value and the fiducial reference it is $g \simeq (0.47$ – $1.5) \times 10^{-7}$ and $m_\psi \simeq 3.4$ – 11×10^{10} GeV, widening to $g \simeq 1.5 \times 10^{-8}$ and $m_\psi \simeq 1.1 \times 10^{10}$ GeV if the two-order non-perturbative enhancement of the last entry is realized; the exact-background computation of Appendix D0 multiplies this entire window by 1.48 (the $T_{\text{reh}}^{-1/2}$ scaling is unchanged), giving $g \simeq (0.7$ – $2.2) \times 10^{-7}$ and $m_\psi \simeq (5.0$ – $16) \times 10^{10}$ GeV across the window; at the model’s own $T_{\text{reh}} = 2.1 \times 10^8$ GeV the matched coupling is $g \simeq 1.5 \times 10^{-7}$ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV).

Concretely, the computation needed to close the loop is the same action with $m_\chi/H_{\text{inf}} \simeq 1.99$, $\xi = 11.1$, $\lambda_0 \simeq 6.70 \times 10^{-8}$ and $g \sim 10^{-7}$, evolved from the end of slow roll through the condensate phase, with two required outputs: the absolute normalization of n_ψ including the oscillating-condensate contribution, and the partition

Table 2 Propagation of the reheating-temperature uncertainty through the locked- N matching and the light-branch abundance relation; computed by `scripts/treh_error_band.py` on the transition-only spectrum (the instantaneous-splice baseline). The inflationary observables are insensitive to T_{reh} , whereas the dark-matter matching point is not and is best quoted as a window. The second row is the model’s own physical- α_s anomaly value. Free-streaming lengths use the diagnostic of this section. The exact homogeneous-background computation of Appendix D0 multiplies the quoted g and m_ψ columns uniformly by 1.48, leaving the $T_{\text{reh}}^{-1/2}$ scaling and the free-streaming conclusion unchanged.

T_{reh} [GeV]	N	n_s	r	g	m_ψ [GeV]	λ_{fs} [Mpc]
1.0×10^8	49.98	0.9616	0.00425	1.5×10^{-7}	1.1×10^{11}	7.8×10^{-20}
2.1×10^8	50.23	0.9618	0.00421	1.0×10^{-7}	7.5×10^{10}	9.5×10^{-20}
1.0×10^9	50.74	0.9621	0.00413	4.7×10^{-8}	3.4×10^{10}	1.4×10^{-19}
1.0×10^{10}	51.51	0.9627	0.00402	1.5×10^{-8}	1.1×10^{10}	2.7×10^{-19}

of the condensate energy between radiation and the stable ψ component (Discussion #3). The first output is now supplied by the exact homogeneous-background mode integration of Appendix D0 ($g \simeq 1.5 \times 10^{-7}$ at the model’s own T_{reh} , validated by the convergence/unitarity/conformal battery). The second requires no additional computation in the regime established there: ψ is a test field with $\rho_\psi/\rho_{\text{end}} \sim 10^{-19}$ at the matching point, so ψ production does not compete with reheating and the partition is fixed by the independent anomaly channel. Two features made the homogeneous computation non-standard but bounded: the condensate fragments inefficiently ($\lambda_{\text{self}} \sim 4 \times 10^{-11}$), so the homogeneous background is a sound starting point; and the fermion coupling is large enough—the adiabaticity parameter (30) being $\sim 3 \times 10^3$ —that no perturbative treatment of ψ production is adequate (the Bogoliubov integration of the mode equation is used instead). What a 3D lattice simulation would still decide is whether the non-linear preheating dynamics—spatial fragmentation, resonance, backreaction of the produced particles—invalidates the homogeneous starting point; parametrically it should not, and the fermion remains the one sector of this framework for which such a non-perturbative cross-check is worth having.

One exact relation between the dark sector and the inflationary scale.

Because $m_\psi = g\Phi_0$ and $\Phi_0 = M_{\text{Pl}}/\sqrt{\xi}$, the dark-matter mass in units of the inflationary Hubble scale is not a free ratio but is fixed by (ξ, g) alone:

$$\frac{m_\psi}{H_{\text{inf}}} = \frac{g}{\sqrt{\xi}} \frac{M_{\text{Pl}}}{H_{\text{inf}}}, \quad (33)$$

which reproduces the transition-only baseline value $m_\psi/H_{\text{inf}} = 4.6 \times 10^{-3}$ at $g = 1.0 \times 10^{-7}$ to machine precision (Table 2). Equation (33) is what makes the dark sector share a scale with inflation rather than merely a field: once ξ and g are fixed, the ratio of the two masses is fixed, and a measurement of either end constrains the other.

5.6 Parallel option: topological defects

Gravitational freeze-in of a Yukawa fermion is not the only way this scalar sector can source dark matter. Non-perturbative $\mathbb{Z}_2$ -breaking dynamics during preheating can leave **topological defects** (domain-wall remnants, textures, or related configurations) with approximately conserved defect energy E_{def} and comoving number, so that

$$n_{\text{def}} \propto a^{-3} \implies \rho_{\text{DM}}^{\text{def}} = E_{\text{def}} n_{\text{def}} \propto a^{-3}, \quad (34)$$

which is the required dust-like scaling, familiar from defect dark-matter scenarios [44]. This channel is a legitimate phenomenological alternative consistent with the same field content. It does *not* supply a first-principles Ω_{DM} without lattice input (formation efficiency, bias, and survival). We therefore list defect dark matter as a parallel channel, not as a completed calculation.

In contrast, a *homogeneous* condensate cannot play this role. From $\ddot{\Phi} + 3H\dot{\Phi} + V' \simeq 0$ near a quadratic minimum one has $\dot{\Phi} \propto a^{-3}$ and hence kinetic energy $\propto a^{-6}$, far too fast to remain a 27% component today; the condensate must decay to reheat the universe (Sec. IV). The same a^{-6} argument underlies the exclusion of single-field dynamical dark energy in Sec. VI.

5.7 Compatibility with reheating

The energy density stored in ψ at production is $\rho_{\psi}(a_{\text{end}}) = m_{\psi} n_{\psi} \simeq 9.5 \times 10^{44} \text{ GeV}^4$ at the fiducial $T_{\text{reh}} = 10^9 \text{ GeV}$ (it scales as T_{reh}^{-1} at fixed Ω_{ψ}), i.e. $\rho_{\psi}/\rho_{\text{end}} \simeq 5.8 \times 10^{-19}$, so ψ does not contest the condensate's dominance; reheating proceeds through the conformal-anomaly decay channel (iv) of Sec. IV, which provides $T_{\text{reh}} \lesssim 10^9 \text{ GeV}$; the initial gravitational pulse supplies only a seed bath and cannot reheat the universe on its own (Sec. IV). Requiring $\Omega_{\psi} = 0.265$ fixes $\rho_{\psi}/\rho_{\text{rad}} \simeq 5.8 \times 10^{-19}$ at reheating (the same number, since $\rho_{\text{rad}}(T_{\text{reh}}) = \rho_{\text{cond}}(a_{\text{reh}})$), so ψ is a spectator to the thermal history and BBN is unaffected. The condensate and ψ thus have disjoint fates—the condensate decays to reheat, ψ survives as dark matter—resolving the tension that would arise if the condensate itself had to be both the reheating fuel and the dark matter.

6 Late-time dark energy: frozen Λ CDM

The dark-energy sector of this model is a calibrated constant, and it is worth stating at the outset why it is not dynamical, since the opposite is the more natural expectation in a single-field framework. The question is decidable within the model itself. If the late-time acceleration came from residual motion of the same field Φ , the condensate would have to survive to the present epoch; it does not. An absolutely stable condensate would overclose the universe by $\sim 10^{18}$ relative to the dark-matter budget, and for any decay rate above $\Gamma \simeq 5 \times 10^{-22} \text{ GeV}$ (that is, $T_{\text{reh}} \gtrsim 10^{-2} \text{ GeV}$) the residual energy density at a_0 underflows to zero with $\Gamma t_0 \gg 1$ (App. D5). Single-field quintessence is excluded kinematically as well: $m_{\chi}/H_0 \sim 10^{55}$, so χ is frozen rather than rolling, and the Klein–Gordon oscillation of χ would drive $w \rightarrow 0$ rather than $w \rightarrow -1$. Dark

energy in this framework is therefore necessarily a constant term of the potential, and $w = -1$ is a derived result rather than an assumption. The price is that V_c is an empirical input fixed by $\Omega_\Lambda = 0.683$, and the 10^{110} hierarchy receives no dynamical explanation (see the fine-tuning discussion below).

6.1 Late-time behavior

Expanding $V_E(\varphi)$ near $\varphi \rightarrow 0$: $V_E(\varphi) \simeq (\lambda_0 M_{\text{Pl}}^4/\xi^2)\varphi^2 + V_c(1 - 4\varphi + 8\varphi^2 + \dots)$. The canonical-field mass is Eq. (12), $m_\chi \sim 3.25 \times 10^{13}$ GeV (locked $N = 50$). With $m_\chi \gg H_0 \sim 10^{-42}$ GeV, the field is underdamped and frozen at the minimum.

The post-inflationary condensate decays during preheating through conformal-anomaly channel (Sec. IV), completing reheating; it does *not* survive as dark matter. The cold dark matter is instead the Dirac fermion ψ of Sec. V, gravitationally produced and redshifting as a^{-3} , a separate $w = 0$ fluid decoupled from both the condensate and the radiation bath.

Once reheating ends, the field settles at $\langle \Phi \rangle = \Phi_0$ and $V_J(\Phi_0) = V_c$ persists as a cosmological constant. The V_c contribution alone gives a cosmological-constant equation of state,

$$w_0 = -1, \quad w(z) = -1 \quad (\forall z), \quad (35)$$

up to residual-condensate corrections bounded in App. D5 (the dominant correction $\Delta w \leq 10^{-10}$ for $T_{\text{reh}} \geq 1$ GeV, consistent with our gravitational-reheating estimate). The $w = -1$ result follows because V_c is the constant part of the potential: it contributes $T_{\mu\nu}^{(\text{CC})} = -V_c g_{\mu\nu}$ and drops out of the scalar equation of motion $\ddot{\chi} + 3H\dot{\chi} + m_\chi^2\chi = 0$, so it evolves independently of both the decaying condensate and the stable ψ fluid. The ψ dark matter is a separate $w = 0$ cold-dark-matter fluid (Sec. V); together with the V_c cosmological constant the late-time model is effectively Λ CDM to extraordinary precision (residual corrections bounded in App. D5; for $T_{\text{reh}} \geq 1$ GeV these are $\leq 10^{-10}$), with $w_{\text{DE}} = -1 + \Delta w$, $|\Delta w| \ll 1$, and no evolution or phantom crossing possible.

Structure growth.

The tree-level conformal decoupling ($g_{\chi ff} = 0$ for ψ , Sec. II E) implies that scalar-field perturbations $\delta\chi$ do not source matter density perturbations in the ψ sector; for SM matter the residual couplings are tiny. The growth equation $\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G\rho_m\delta_m = 0$ is identical to Λ CDM at leading order. The effective coupling induced by graviton loops ($\sim 10^{-11}g$, see loop discussion in Sec. II E) is far too weak to produce a measurable deviation in $f\sigma_8$ ($\Delta f/f \sim 10^{-22}$). The model therefore predicts $f\sigma_8(z)$ indistinguishable from Λ CDM and automatically compatible with DESI, eBOSS, and future Stage-IV surveys.

6.2 Residual motion bound

Three sources of residual χ motion that could perturb the frozen minimum are bounded (full derivations in App. D5):

- **Ricci driving:** $\Delta w \sim (H_0/m_\chi)^2 \sim 2 \times 10^{-111}$.

- **Quantum fluctuations:** $\Delta w \sim H_0^4/V_c \sim 2 \times 10^{-121}$.
- **Residual condensate tail:** the oscillating condensate redshifts as a^{-3} (pressureless dust, $w = 0$), not as dark energy. Its relic density is subject to the overclosure constraint $\rho_{\text{cond}}(a_0) \leq \rho_{\text{crit}}$, i.e. $\rho_{\text{cond}}(a_0)/V_c \leq 1/\Omega_\Lambda$ (or $\leq \Omega_{\text{DM}}/\Omega_\Lambda \simeq 0.39$ if the condensate is required not to exceed the observed dark-matter budget). This bound is *satisfied*, but only because the condensate decays: an absolutely stable condensate would give $\rho_{\text{cond}}(a_0)/V_c \simeq 7 \times 10^{17}$, i.e. it would overclose the universe by $\sim 10^{18}$. What makes the bound hold is the decay-completeness condition $\Gamma t_0 \gg 1$, satisfied by an enormous margin for any $T_{\text{reh}} \gtrsim 10^{-2}$ GeV (e.g. $\Gamma t_0 \simeq 3 \times 10^{20}$ at the BBN floor and $\simeq 3 \times 10^{42}$ at $T_{\text{reh}} = 10^9$ GeV), so that the residual at a_0 is strictly zero. A dedicated integration of the two-fluid Boltzmann equations (App. D5c) confirms this: tracking the condensate as $Y \equiv \rho_{\text{cond}} a^3$ and the radiation as $R \equiv \rho_{\text{rad}} a^4$, the residual today vanishes for every $\Gamma \neq 0$ across $T_{\text{reh}} \in [10^{-2}, 10^{15}]$ GeV. The condensate therefore contributes to Ω_m only through its decay products, not to Δw_{DE} .

The first two are far below the DESI/Euclid 3σ sensitivity of $\Delta w \sim 10^{-3}$; the third is bounded by the overclosure constraint and the rapid damping $\tau \sim m_\chi^{-1} \sim 10^{-38}$ s (App. D4), which jointly render the condensate cosmologically irrelevant at late times.

The condensate decays completely during preheating, so it leaves no residual component that could perturb the frozen minimum. The dark matter ψ is a separate $w = 0$ cold-dark-matter fluid and does not enter the dark-energy equation of state: $w_{\text{DE}} = -1$ up to residual-condensate corrections bounded in App. D5 (negligible for $T_{\text{reh}} \geq 1$ GeV). The corresponding total effective equation of state $w_{\text{tot}} = -V_c/(\rho_\psi + V_c)$ is *not* the quantity DESI/BAO/SNe constrain—standard Λ CDM has $w_{\text{tot}} \approx -0.69$ and is likewise unexcluded.

6.3 Calibration and fine-tuning

V_c is calibrated by $\Omega_\Lambda = 0.683$:

$$V_c = \Omega_\Lambda \cdot 3H_0^2 M_{\text{Pl}}^2 \approx 2.5 \times 10^{-47} \text{ GeV}^4. \quad (36)$$

The 10^{110} hierarchy between $V_0 \sim 10^{63} \text{ GeV}^4$ and V_c is spanned by the conformal factor $e^{-4\varphi}$ but *not* dynamically explained; this is the cosmological constant problem, shared with all Λ CDM models. This model reproduces a late-time cosmological constant with $w = -1$ (up to corrections $\leq 10^{-10}$ for $T_{\text{reh}} \geq 1$ GeV; see App. D5), but does not address the core fine-tuning puzzles. Two distinct tuning problems remain: (i) why $\Phi_0 = M_{\text{Pl}}/\sqrt{\xi} \sim 10^{17}$ GeV sets the Planck/electroweak hierarchy—the VEV is a fixed point of the potential by construction but its magnitude is not dynamically explained; (ii) why $V_c \sim 10^{-47} \text{ GeV}^4$ is 10^{110} times smaller than the inflationary scale $V_0 \sim 10^{63} \text{ GeV}^4$ —the conformal factor $e^{-4\varphi}$ maps V_c from the Jordan to the Einstein frame but the tiny value of V_c itself is not derived from a symmetry or dynamical principle [34]. Moreover, the heavy fields χ ($m_\chi \sim 10^{13}$ GeV) and ψ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV) contribute radiative corrections $\delta V_c \sim m^4/(16\pi^2) \sim 10^{42}\text{--}10^{52} \text{ GeV}^4$, exacerbating the fine-tuning by an additional $\sim 10^{89}\text{--}10^{98}$ orders relative to the observed V_c ; this makes the cosmological constant problem quantitatively more severe in this model

than in the Standard Model alone, an issue shared with any theory containing heavy scalar or fermion degrees of freedom and deferred to a UV completion. V_c is an empirical observational input, not dynamically derived. During reheating, the minimum at $\Phi = \Phi_0$ is protected against thermal and curvature perturbations: the field mass $m_\Phi \sim 2.7 \times 10^{14}$ GeV exceeds the reheating temperature $T_{\text{reh}} \sim 10^9$ GeV by five orders of magnitude, thermal mass corrections $\delta m_\Phi^2 \sim g^2 T^2$ are $\leq 10^{-20} m_\Phi^2$, and curvature-driven displacements $\sim \xi \Phi_0 R / m_\Phi^2$ are $\leq 10^{-13}$; the minimum remains stable throughout all post-inflationary epochs. A detection of $w_0 \neq -1$ would exclude this model and Λ CDM alike, and the latest data lean that way: DESI DR2, combined with the CMB, prefers time-evolving dark energy over Λ CDM at 3.1σ (2.8–4.2 σ once supernovae are included), in the quadrant $w_0 > -1$, $w_a < 0$ [36]. The present model has no degree of freedom that can produce such an evolution— χ is frozen and V_c is a constant—so its dark-energy sector is excluded if that preference survives as a physical effect rather than as a systematic. Because Λ CDM predicts the same $w_0 = -1$, the test bears on the cosmological-constant hypothesis rather than on the unification this framework proposes.

6.4 Drift of the fundamental constants

The same residual Ricci-driven displacement that enters the Δw budget above also drives a drift of the fundamental constants, providing a consistency check that the induced-gravity framework generates no observable violation of the equivalence principle or time-variation of couplings. All effects derived here lie > 90 orders below current experimental sensitivity; this subsection may be skipped on a first reading without loss of continuity.

In FRW, the linearized field equation reads

$$\delta\ddot{\Phi} + 3H\delta\dot{\Phi} + (2\lambda_0\Phi_0^2 - \xi R)\delta\Phi = \xi\Phi_0 R. \quad (37)$$

In the quasi-static limit:

$$\frac{\Delta\mathcal{A}}{\mathcal{A}} = +\frac{\xi R}{\lambda_0\Phi_0^2}. \quad (38)$$

With $R_0 \approx 9.2H_0^2$: $\Delta\mathcal{A}/\mathcal{A}|_0 \approx +6.3 \times 10^{-11}$, far below foreseeable sensitivity.

Fine-structure-constant drift.

$\Delta\alpha/\alpha = (de/d\Phi)(\Phi_0/e) \cdot \Delta\mathcal{A}/\mathcal{A}$. Three scenarios: (i) e independent of $\Phi \Rightarrow \Delta\alpha/\alpha = 0$; (ii) mass-coupling-dominance $\Rightarrow |\Delta\alpha/\alpha| \sim 10^{-110}$; (iii) logarithmic dependence $\Rightarrow$ even smaller. Scenario (ii) lies ~ 90 orders below current sensitivity.

7 Standard Model embedding and renormalization-group stability

The analysis of Sec. II E demonstrates that the direct Yukawa coupling $-g\Phi\bar{\psi}\psi$ is exactly decoupled by the conformal identity $\Omega = \Phi/\Phi_0$. This section constructs the minimal SM extension— Φ couples to SM fields only through the metric (no direct Higgs portal)—and verifies RG stability.

7.1 Jordan-frame action with Standard Model fields

The total Jordan-frame action is the sum of the scalar-tensor sector (1) and the SM Lagrangian in curved spacetime:

$$S_{\text{SM}} = \int d^4x \sqrt{-g} \left[-\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \bar{\psi}_f i\gamma^\mu D_\mu \psi_f + |D_\mu H|^2 - V_{\text{SM}}(H) - y_f \bar{\psi}_L \psi_R H + \text{h.c.} \right], \quad (39)$$

where $F_{\mu\nu}^a$ are SM gauge field strengths, ψ_f denotes SM fermion families, H is the Higgs doublet with $V_{\text{SM}}(H) = -\mu_H^2 |H|^2 + \lambda_H |H|^4$, and y_f are the Yukawa couplings. Φ is a gauge singlet with no renormalizable direct coupling to SM fields; setting the Higgs portal $\lambda_{\Phi H} \Phi^2 |H|^2$ to zero is technically natural [11] (the limit $\lambda_{\Phi H} \rightarrow 0$ enhances the symmetry by decoupling Φ from the Higgs sector). Radiative corrections induce $\delta\lambda_{\Phi H}$ only through graviton exchange, giving $\delta\lambda_{\Phi H} \sim \xi\lambda_0/(16\pi^2) \times (m_\Phi^2/M_{\text{Pl}}^2) \simeq 6 \times 10^{-17}$ (with $m_\Phi \simeq 2.7 \times 10^{14}$ GeV), far below any observable threshold. All SM matter couples to Φ only through the Jordan-frame metric.

7.2 Gauge sector

Under $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$, the gauge kinetic term transforms as

$$-\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} \sqrt{-g} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} \sqrt{-\tilde{g}}, \quad (40)$$

because $g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}$ is conformally invariant in four dimensions. Gauge fields are not rescaled; SM gauge couplings g_1, g_2, g_3 are unchanged and there is no tree-level χ -gauge vertex. At the 1-loop level, however, the conformal (trace) anomaly generates an effective χ -gauge coupling $\mathcal{L}_{\text{eff}} \supset (b_i \alpha_i / 4\pi) (\chi / (M_{\text{Pl}} \sqrt{6 + 1/\xi})) F_i^2$, opening the $\chi \rightarrow gg, \gamma\gamma$ decay channels at $\Gamma_{\text{anom}} \sim \mathcal{O}(10^{-1})$ GeV. Because this is a continuous perturbative decay it is the channel that actually reheats the universe, giving the fiducial $T_{\text{reh}} \lesssim 10^9$ GeV (Sec. IV); the gravitational pulse (i) is a single initial burst that dilutes faster than the condensate and cannot do so on its own. The tree-level decoupling of χ from gauge bosons remains exact for the purpose of fifth-force bounds, which are set by the scalar's large mass.

Massive gauge bosons.

After electroweak symmetry breaking, the $W^\pm$ and Z bosons acquire mass. Under the conformal transformation $g_{\mu\nu} = \Omega^{-2} \tilde{g}_{\mu\nu}$, the mass term transforms as $\frac{1}{2} m_W^2 g_{\mu\nu} W^\mu W^\nu \sqrt{-g} \rightarrow \frac{1}{2} (m_W / \Omega)^2 \tilde{g}_{\mu\nu} \tilde{W}^\mu \tilde{W}^\nu \sqrt{-\tilde{g}}$ (with $\tilde{W}_\mu \equiv \Omega^{-1} W_\mu$), so the effective Einstein-frame mass is $\tilde{m}_W = m_W / \Omega$. Since $\Omega = \Phi / \Phi_0 \rightarrow 1$ after reheating (the field settles at $\Phi = \Phi_0$ with residual displacement $\delta\Phi / \Phi_0 \sim 10^{-111}$; Sec. VI), the standard W and Z masses are recovered to extraordinary precision in the late-time SM. During inflation, m_W is exponentially suppressed (as is the entire electroweak scale), but this is irrelevant since the Higgs field is displaced from its minimum and electroweak symmetry is not yet broken.

7.3 Fermion sector

The curved-space fermion kinetic term is conformally invariant after $\psi = \Omega^{3/2}\tilde{\psi}$ [23]. Higgs Yukawa terms transform as

$$y_f \bar{\psi}_L \psi_R H \sqrt{-g} \longrightarrow y_f \Omega^{3/2} \tilde{\psi}_L \Omega^{3/2} \tilde{\psi}_R (\tilde{H}/\Omega) \Omega^{-4} \sqrt{-\tilde{g}} = y_f \Omega^{-2} \tilde{\psi}_L \tilde{\psi}_R \tilde{H} \sqrt{-\tilde{g}}, \quad (41)$$

where $\tilde{H} \equiv \Omega H$. Unlike the direct $-g\Phi\bar{\psi}\psi$ coupling, where $\Phi \propto \Omega$ cancels all factors, the SM Higgs Yukawa leaves a residual factor Ω^{-2} . The Yukawa fermion ψ of Sec. V is taken to be a Dirac, gauge-singlet fermion with no Higgs portal ($\bar{L}\tilde{H}\psi$ set to zero, technically natural [11]); it is therefore distinct from the SM fermions ψ_f treated here, has no residual χ coupling, and is stable under the discrete $\mathbb{Z}_2^\psi$ symmetry (Sec. V). The physical Einstein-frame fermion mass is $m_f = m_f^{(0)}/\Omega = m_f^{(0)}e^{-\varphi}$: the Ω^{-2} of Eq. (41) multiplies the field $\tilde{H}$, whose expectation value $\langle \tilde{H} \rangle = \Omega v$ carries a compensating Ω . Hence the effective χ -fermion coupling is

$$g_{\chi ff} \simeq \frac{m_f^{(0)}}{M_{\text{Pl}}\sqrt{6+1/\xi}} \sim 10^{-22}\text{--}10^{-17}, \quad (42)$$

many orders below the Cassini bound [12]. The range $\sim 10^{-22}\text{--}10^{-17}$ reflects the fermion-mass hierarchy: the lower end corresponds to the electron ($m_e \sim 0.5$ MeV) and the upper end to the top quark ($m_t \sim 173$ GeV); even at the top end the coupling is far below the solar-system bound, and the corresponding decay is negligible for reheating (see Sec. IV, channel (ii)). The corresponding decay rate $\Gamma(\chi \rightarrow ff)/m_\chi \leq 5 \times 10^{-35}$ (quoted for the top, the largest coupling) is negligible for reheating. For comparison, $\Gamma_{\chi \rightarrow t\bar{t}} \sim 1.5 \times 10^{-21}$ GeV versus the anomaly width $\Gamma_{\text{anom}} \sim \mathcal{O}(10^{-1})$ GeV of channel (iv), a $\sim 7 \times 10^{19} \times$ suppression; the direct χff coupling is therefore completely irrelevant for reheating.

7.4 Higgs sector and field-space metric

After rescaling $H = \tilde{H}/\Omega$, the Einstein-frame kinetic terms for χ and $\tilde{H}$ read

$$\mathcal{L}_{\text{kin}} = \frac{1}{2}(\partial\chi)^2 + \frac{1}{2}e^{-2\varphi}|D_\mu\tilde{H}|^2, \quad \frac{d\varphi}{d\chi} = \frac{1}{M_{\text{Pl}}\sqrt{6+1/\xi}}. \quad (43)$$

At late times $\varphi \rightarrow 0$, the Higgs kinetic term reduces to the canonical SM form. Since φ is frozen with $m_\chi \gg H_0$, all χ - $\tilde{H}$ mixing is kinematically inaccessible; the SM is recovered exactly.

During inflation ($\varphi \gg 1$), the Higgs mass term is exponentially suppressed by $e^{-4\varphi}$, so H is effectively massless and does not affect inflationary dynamics. Isocurvature perturbations from Higgs- Φ field-space curvature are estimated to be well below Planck sensitivity for $\varphi \geq 5$ in generic Higgs-inflation setups [37]; a dedicated computation of the isocurvature power spectrum for the present model's specific parameters ($\xi = 11.1$, $\lambda_0 \simeq 6.70 \times 10^{-8}$ at locked $N = 50$, $\xi_h = 0$) and a direct comparison

with Planck 2018 isocurvature bounds [16] have not been performed and are deferred to a future multifield analysis. A Higgs non-minimal coupling $\xi_h |H|^2 R$ is in principle allowed in the Jordan frame; if present, it would induce an Einstein-frame term $\xi_h |\tilde{H}|^2 \tilde{R}/\Omega^2$, contributing to the field-space metric during inflation. For simplicity we set $\xi_h = 0$; a nonzero $\xi_h \sim \mathcal{O}(1)$ would shift the inflationary observables by $\leq 10^{-4}$ [3] and does not affect our conclusions.

Electroweak phase transition.

Since $T_{\text{reh}} \sim 10^9 \text{ GeV} \gg v$, the electroweak symmetry is restored after reheating and broken when the Universe cools below $T \sim 100 \text{ GeV}$. With $\xi_h = \lambda_{\Phi H} = 0$, the Φ -Higgs mixing vanishes at tree level and the electroweak phase transition proceeds as in the SM (a smooth crossover [38]), unmodified by the inflaton background. If loop effects were to induce a small $\lambda_{\Phi H}$, the transition dynamics could in principle be altered; a quantitative study of electroweak baryogenesis in this framework is deferred to future work.

7.5 Renormalization-group stability of ξ and λ_0

For a real scalar singlet with a Yukawa fermion ψ , the 1-loop β -functions in the $\overline{\text{MS}}$ scheme are [22, 39]

$$\beta_{\lambda_0} \equiv \frac{d\lambda_0}{d\ln\mu} = \frac{9\lambda_0^2}{8\pi^2} + \mathcal{O}(\lambda_0^3), \quad (44)$$

$$\beta_\xi \equiv \frac{d\xi}{d\ln\mu} = \left(\xi - \frac{1}{6}\right) \frac{1}{16\pi^2} (3\lambda_0 + g^2) + \mathcal{O}(\lambda_0^2, g^3), \quad (45)$$

where $\gamma_\Phi = g^2/(16\pi^2) + \mathcal{O}(\lambda_0^2)$ is the field anomalous dimension (driven at 1-loop only by the Yukawa, since the quartic contributes to the wavefunction renormalization first at 2-loop). The coefficient $9/(8\pi^2)$ in β_{λ_0} is the standard 1-loop result for the convention $V \supset (\lambda_0/4)\Phi^4$ (equivalently $\lambda_{\text{canon}} = 6\lambda_0$ in $V \supset \lambda_{\text{canon}}\Phi^4/4!$, for which $\beta_{\lambda_{\text{canon}}} = 3\lambda_{\text{canon}}^2/16\pi^2$). In β_ξ there are *two* distinct 1-loop drives, both proportional to $(\xi - \frac{1}{6})$: (a) the multiplicative term $(\xi - \frac{1}{6})\gamma_\Phi$ from the Yukawa loop, and (b) a *direct* curvature-coupling renormalization $(\xi - \frac{1}{6})3\lambda_0/(16\pi^2)$ from the quartic-curvature cross term in the 1-loop effective action [23, 39]. The latter does not pass through γ_Φ (the quartic does not renormalize the wavefunction at 1-loop, as the tadpole integrates to zero in dimensional regularization) but renormalizes the curvature operator $\Phi^2 R$ directly; it is the same structure that makes the Higgs quartic drive β_ξ in Higgs inflation. We retain both terms.

Numerically, $\Delta\lambda_0 \leq (9\lambda_0^2/8\pi^2) \times 60 \sim 3 \times 10^{-14}$, completely negligible. For β_ξ , the two pieces over the inflationary-to-low-energy range ($\Delta\ln\mu \sim 60$) are

$$\Delta\xi_{\lambda_0} \sim \left(\xi - \frac{1}{6}\right) \frac{3\lambda_0}{16\pi^2} \times 60 \sim 8.5 \times 10^{-7}, \quad (46)$$

$$\Delta\xi_g \sim (\xi - \tfrac{1}{6}) \frac{g^2}{16\pi^2} \times 60 \sim \begin{cases} 9.1 \times 10^{-14} & (g = 1.5 \times 10^{-7}, \text{ abundance-matched}) \\ 2.2 \times 10^{-9} & (g = 2.3 \times 10^{-5}) \\ 4.2 \times 10^{-8} & (g = 10^{-4}) \end{cases} \quad (47)$$

so the quartic drive dominates the Yukawa drive by $\sim 9 \times 10^6$ at the abundance-matched coupling $g \simeq 1.5 \times 10^{-7}$ (and by $\sim 20\text{--}380\times$ at the larger g of the exponential framework). Both contributions carry the same sign (the curvature-coupling renormalization is additive [23, 39]), so no cancellation occurs. At the fiducial $\xi = 11.1$, the total $\Delta\xi \leq 10^{-6}$ remains utterly negligible; across the full scan range $\xi \in [1, 100]$, the prefactor $(\xi - \frac{1}{6})$ varies between 0.83 and 99.8, giving $\Delta\xi(\xi) \in [6.5 \times 10^{-8}, 7.8 \times 10^{-6}]$, still $\leq 10^{-5}$. At 2-loop, $\gamma_\Phi \sim \lambda_0^2/(16\pi^2)^2 \sim \mathcal{O}(10^{-19})$ is subdominant.

The parameters $\xi \simeq 11.1$ and $\lambda_0 \simeq 6.70 \times 10^{-8}$ (locked $N = 50$; 6.45×10^{-8} at $N = 51$) are therefore stable under RG flow: no fine-tuned cancellation is required. We note that the small $g \simeq 1.5 \times 10^{-7}$ required for dark matter (whose precise value depends on T_{reh} ; see Sec. V) makes the Yukawa contribution to β_ξ even smaller, but the quartic drive $\sim 8.5 \times 10^{-7}$ —which is g -independent—sets the floor of the running; a much larger quartic would be needed to spoil the stability, and $\lambda_0 \sim 10^{-7}$ is far below that threshold. Since $\beta_{\lambda_0} \propto \lambda_0^2 > 0$, the self-coupling decreases monotonically from the UV to the IR; with the total shift $\Delta\lambda_0 \sim 3 \times 10^{-14}$ over $\Delta\ln\mu \sim 60$, λ_0 remains positive at all scales between the inflationary scale and the electroweak scale—no vacuum instability arises in the Φ sector. (The known SM Higgs vacuum stability problem [40] is unaffected by the present model since $\lambda_{\Phi H} = 0$ decouples the two sectors.)

Quantum stability of the VEV.

The 1-loop Coleman–Weinberg correction shifts the VEV by $\delta\Phi_0/\Phi_0 \sim \lambda_0/(16\pi^2) \sim 4 \times 10^{-10}$, corresponding to a fractional change in the effective Planck mass $\delta M_{\text{Pl}}/M_{\text{Pl}} = \delta\Phi_0/\Phi_0$ of the same order—far below the 10^{-4} precision of the Newton constant measurement [41]. The VEV is therefore stable under quantum corrections without additional fine-tuning. The heavy fields χ ($m_\chi \sim 10^{13}$ GeV) and ψ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV) enter the effective potential at 1-loop but their contributions are suppressed by the ratio of their masses to the UV cutoff, and the tree-level VEV is protected by the shift symmetry of the potential’s construction $V'_J(\Phi_0) = 0$. The hierarchy $m_{\chi,\psi} \ll \Lambda_J \sim M_{\text{Pl}}/\xi \sim 2.2 \times 10^{17}$ GeV ensures that decoupling works as usual: heavy-field threshold corrections to low-energy observables are power-suppressed by $\mathcal{O}(m^2/\Lambda_J^2)$, namely $(m_\chi/\Lambda_J)^2 \simeq 2.2 \times 10^{-8}$ and $(m_\psi/\Lambda_J)^2 \simeq 2.4 \times 10^{-13}$.

Impact on SM Higgs stability.

The heavy fields χ and ψ couple to the SM only through gravity and (for ψ) the discrete $\mathbb{Z}_2^\psi$ gauge sector; they do not carry SM gauge quantum numbers. Their threshold contributions to the Higgs quartic β -function are suppressed by at least two powers of $m_{\chi,\psi}/M_{\text{Pl}} \leq 10^{-5}$ (gravity-mediated) or by the minuscule radiatively induced $\delta\lambda_{\Phi H} \simeq 6 \times 10^{-17}$ (Sec. VII). Consequently, the SM Higgs vacuum stability analysis of Ref. [40] is unaltered at the leading-log level; the metastability boundary for λ_H shifts by

$\leq 10^{-10}$ in the Φ - χ - ψ parameter range. The model therefore neither exacerbates nor alleviates the known SM Higgs vacuum stability problem—the two sectors are RG-decoupled to extraordinary precision.

Finite-temperature stability.

At the reheating temperature $T_{\text{reh}} \sim 10^9$ GeV, thermal corrections to the Φ effective potential are controlled by the coupling of χ to the thermal bath. The χ -SM coupling is gravitational or anomaly-induced, with effective Yukawa couplings $g_{\chi ff} \sim m_f/M_{\text{Pl}} \leq 10^{-17}$ (Eq. (42)); the thermal mass correction is $\delta m_\chi^2(T) \sim g_{\chi ff}^2 T^2 \leq 10^{-14} T^2 \ll m_\chi^2$, completely negligible. The Φ vacuum at $\langle \Phi \rangle = \Phi_0$ remains stable throughout the entire thermal history—no symmetry restoration or thermal destabilization occurs. Similarly, the heavy scalar mass $m_\chi \sim 3.2\text{--}3.3 \times 10^{13}$ GeV is over four orders of magnitude above $T_{\text{reh}} \sim 10^9$ GeV, so χ particles are never thermally populated; the field remains frozen at the minimum during and after reheating.

SM couplings ($g_1, g_2, g_3, \lambda_H, y_t$) follow their conventional β -functions at all scales below $\Lambda_J \sim M_{\text{Pl}}/\xi$; gauge-coupling matching between frames is trivial because the conformal transformation $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$ is a field redefinition that leaves physical S -matrix elements invariant—any apparent frame-dependence of the running is compensated by the rescaling of the renormalization scale $\mu \rightarrow \mu/\Omega$, and since $\Omega \rightarrow 1$ in the late-time universe the SM β -functions are unaffected at all observable epochs. At the quantum level, the conformal anomaly generates the χFF coupling discussed in Sec. IV but does not modify the *running* of the gauge couplings themselves, as the anomaly enters through the trace of the energy-momentum tensor rather than through the β -functions [23].

7.6 Summary of embedding consistency

The minimal SM embedding satisfies all observational constraints:

- **Gauge sector:** conformally invariant at tree level; conformal-anomaly-induced χ -gauge coupling at 1-loop (discussed in Secs. IV and VII) does not affect the T_{reh} or fifth-force conclusions.
- **Fermion sector:** residual χ -fermion coupling $\sim m_f/M_{\text{Pl}}$, automatically satisfying solar-system bounds.
- **Higgs sector:** late-time SM recovered; Higgs- Φ mixing is kinematically inaccessible.
- **RG stability:** λ_0 shifts by $\Delta\lambda_0/\lambda_0 \sim 10^{-6}$ and ξ by $\Delta\xi \leq 10^{-6}$ at the fiducial $\xi = 11.1$ (across $\xi \in [1, 100]$, $\Delta\xi \leq 10^{-5}$; driven at 1-loop by the ψ Yukawa $\propto g^2$ and a direct quartic-curvature term $\propto \lambda_0$, the latter dominating) from the inflationary scale to low energies.

The only substantive open issue is the Higgs field-space metric during inflation (potential isocurvature perturbations); a dedicated multifield analysis is deferred. No fine-tuning is introduced: ξ and λ_0 are RG-stable, and setting $\lambda_{\Phi H} = 0$ is technically natural.

8 $\mathbb{Z}_2$ domain wall: classical EFT applicability boundary

Nature and scope.

This section documents a technical property of the $F = \xi\Phi^2$ coupling: within the classical EFT domain, $\mathbb{Z}_2$ domain walls cannot be treated as thin shells. This is an algebraic no-go statement; cosmologically, $\mathbb{Z}_2$ walls are inflationarily diluted (Sec. II), so it carries no phenomenological consequence and is not an observational test.

8.1 Thin-shell matching degeneracy at $\Phi = 0$

Starting from Eq. (3), projecting onto tangential components and integrating across the shell ($[\Phi] = 0$ required) gives the singular terms. The integrated Lanczos condition is

$$F(\Phi_s)[K_{ab}] = -S_{ab} + \frac{1}{2}h_{ab}S - \frac{1}{2}h_{ab}F'(\Phi_s)[\partial_n\Phi]. \quad (48)$$

At $\Phi_s = 0$, $F(0) = 0$ and $F'(0) = 0$, giving $0 = -S_{ab} + \frac{1}{2}h_{ab}S$. For a tension wall $S_{ab} = -\sigma h_{ab}$, $S = -3\sigma$, so $0 = -\frac{1}{2}\sigma h_{ab} \neq 0$, an inconsistency that no choice of $[K_{ab}]$ can resolve.

8.2 Physical interpretation

The algebraic inconsistency at $\Phi = 0$ is a structural property of $F = \xi\Phi^2$. The surface $\Phi = 0$ lies beyond the classical EFT regime because:

- $F(0) = 0$ makes $G_{\text{eff}} \propto 1/F \rightarrow \infty$ (gravitational coupling diverges).
- $\Omega^2 = \xi\Phi^2/M_{\text{Pl}}^2 \rightarrow 0$ makes the Einstein-frame potential $V_E = V_J/\Omega^4 \rightarrow \infty$ and the conformal transformation singular.

This shows that within the EFT, $\mathbb{Z}_2$ domain walls cannot be treated as classical thin shells: the Israel–Lanczos matching degenerates at $\Phi = 0$ owing to $F(0) = F'(0) = 0$. This is a structural property of the $F = \xi\Phi^2$ coupling, not an observational constraint, as $\mathbb{Z}_2$ walls are inflationarily diluted (Sec. II). The bubble-wall channel ($F(\Phi_0) = M_{\text{Pl}}^2 \neq 0$), which involves field configurations away from $\Phi = 0$ and connects different local vacua via spherical bubbles rather than planar $\mathbb{Z}_2$ walls, is not covered by the $F(0) = 0$ thin-shell degeneracy and is not ruled out by the junction-condition analysis alone. Whether bubble nucleation actually occurs and whether any resulting bubbles are cosmologically benign (evading constraints from gravitational-wave backgrounds, BBN, or CMB anisotropy) depends on the tunneling action S_E , which we have not computed; a full assessment of bubble-wall cosmology in this model is beyond the present scope.

From the Einstein-frame perspective, $\Omega^2 \rightarrow 0$ at $\Phi = 0$ causes $V_E \rightarrow \infty$ and $\sigma_E \rightarrow \infty$. These divergences are conformal-mapping artifacts; the Jordan-frame σ_{wall} remains finite (Eq. (5)). Explicitly, in the Einstein frame the canonically normalized

Fig.3 Domain-wall / EFT boundary at $\Phi = 0$ (two frames)

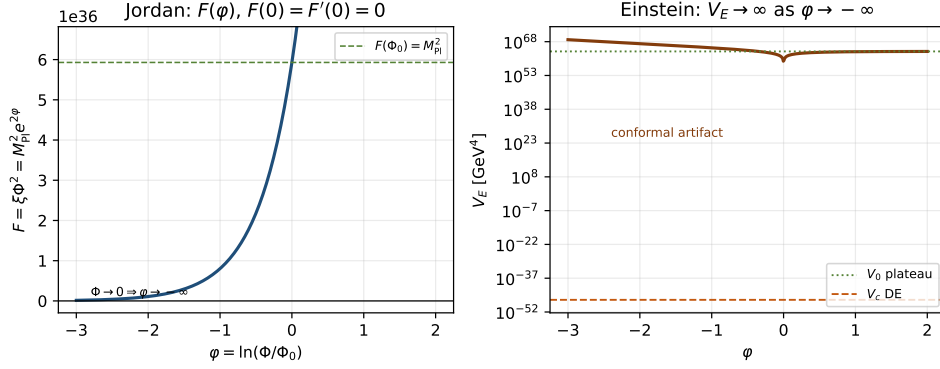


Fig. 3 Two-frame comparison of the $\mathbb{Z}_2$ domain-wall / EFT boundary at $\Phi = 0$. Left: Jordan-frame coupling $F(\varphi) = M_{\text{Pl}}^2 e^{2\varphi}$ with $F \rightarrow 0$ as $\varphi \rightarrow -\infty$ ($\Phi \rightarrow 0$), so $F(0) = F'(0) = 0$ and thin-shell matching degenerates. Right: Einstein-frame V_E diverges in the same limit purely through the conformal map $\Omega \rightarrow 0$; V_0 and calibrated V_c are indicated. Regenerated by `scripts/fig3.domain-wall.py`.

Table 3 Degeneracy of the domain-wall matching condition at $\Phi = 0$ in the two conformal frames.

Frame	Degeneracy at $\Phi = 0$	Origin
Jordan	$F(0) = F'(0) = 0, 0 = -\frac{1}{2}\sigma h_{ab}$	Exact algebraic identity
Einstein	$V_E \rightarrow \infty, \sigma_E$ divergent	Conformal-mapping artifact

field χ satisfies $d\chi/d\varphi = M_{\text{Pl}}\sqrt{6 + 1/\xi}$, and the wall tension integral becomes

$$\sigma_E = \int d\chi \sqrt{2V_E(\chi)} = M_{\text{Pl}} \sqrt{6 + \frac{1}{\xi}} \int d\varphi \sqrt{2V_E(\varphi)}, \quad (49)$$

with

$$\sqrt{2V_E(\varphi)} \xrightarrow{\varphi \rightarrow -\infty} M_{\text{Pl}}^2 \sqrt{\frac{\lambda_0}{2\xi^2}} e^{-2\varphi}, \quad (50)$$

so $\sigma_E \propto \int_{-\infty}^{\infty} e^{-2\varphi} d\varphi$ diverges at the $\Phi = 0$ boundary ($\varphi \rightarrow -\infty$)—a pure conformal-singularity artifact of the $\Omega \rightarrow 0$ limit. The divergence is driven by the quartic term in V_E ; crucially, the $V_c e^{-4\varphi}$ contribution is $\sim 10^{110}$ times smaller than the quartic-driven barrier at $\varphi \rightarrow -\infty$ and is irrelevant for the wall dynamics. The Jordan-frame tension remains the physically meaningful finite quantity.

9 The unified picture

The framework presented in this paper achieves a unified description of the major cosmological sectors through a single scalar field Φ and a single coupling $F = \xi\Phi^2$; the dark matter is a Dirac fermion ψ already coupled to Φ by the Yukawa term, whose mass

Table 4 Summary of the unified framework: each phenomenon emerges from a different regime of the same field Φ and the same coupling $F = \xi\Phi^2$; the dark matter is a Dirac fermion ψ whose mass $m_\psi = g\Phi_0$ originates from the same VEV. The relic abundance is matched on the light branch ($g \simeq 1.5 \times 10^{-7}$); its absolute normalization is fixed at the homogeneous level (Appendix D0), with a full preheating simulation as the remaining cross-check.

Phenomenon	Field regime	Mechanism	Observable
Planck mass	$\langle\Phi\rangle = \Phi_0$	Induced gravity	G_N
Inflation	$\Phi \gg \Phi_0$	Starobinsky plateau	n_s, r
Reheating	$\Phi \sim \Phi_0$	Anomaly	T_{reh}
DM (cand.)	ψ Dirac / defects	Grav. production; topological defects	Ω_{DM} at $g \simeq 10^{-7}$
Dark energy	$\Phi = \Phi_0$ frozen	$V_E(0) = V_c$	$w_0 = -1$
Fifth force	$\Omega = \Phi/\Phi_0; m_\chi$	Conformal decoupling	$\gamma = 1$
$\mathbb{Z}_2$ wall	$\Phi = 0$	$F = F' = 0$ EFT bound.	Classical boundary
SM	$\Phi = \Phi_0$	$\chi f \bar{f} \sim m_f/M_{\text{Pl}}$	Standard couplings

$m_\psi = g\Phi_0$ originates from the same VEV. Table 4 summarizes the correspondence between field regimes and emergent phenomena.

All entries except dark matter are established at the level of rigorous field theory; the dark matter entry is a structurally stable Dirac fermion (under the assumed discrete gauge UV completion) whose relic abundance is matched on the light branch at $g \simeq 1.5 \times 10^{-7}$ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV) by the exact homogeneous-background mode integration (Appendix D0), with the non-perturbative (lattice) cross-check of the homogeneous assumption remaining open.

Two structural features operate in distinct sectors: (a) the conformal identity $\Omega = \Phi/\Phi_0$ (a consequence of $F = \xi\Phi^2$) decouples the scalar from the dark-matter fermion ψ at tree level, closing the perturbative Yukawa reheating channel; (b) for Standard Model matter, whose masses arise via the Higgs mechanism, the residual χ coupling $\sim m_f/M_{\text{Pl}}$ is suppressed by the scalar’s large mass $m_\chi \sim 10^{13}$ GeV (Yukawa range $\sim 10^{-27}$ cm), satisfying the Cassini bound without a chameleon mechanism.

10 Testability and falsifiable predictions

a. Level 1: Observational predictions (rigorous).

- Inflationary observables (locked N):** exact potential slow roll with $N = \ln(a_{\text{end}}/a_*)$. A measurement $r > 0.01$ at 95% CL by LiteBIRD [18] excludes the model: Planck 2σ allows $N \approx 45\text{--}56$ where $r \leq 0.0052$ (the statistical n_s band, capped at $N_{\text{max}} \simeq 55.6$ by instantaneous reheating; Table 1).
- Dark energy equation of state:** $w_0 = -1$ (up to residual-condensate corrections $\leq 10^{-10}$ for $T_{\text{reh}} \geq 1$ GeV; App. D5). DESI DR2 already reports a 3.1σ preference for $w_0 \neq -1$ [36], which this model cannot accommodate (the field is frozen and V_c is a constant); but because Λ CDM predicts the same value, such a detection would exclude both rather than discriminate between them.
- Solar-system fifth-force bounds:** satisfied via the scalar’s large mass ($m_\chi \sim 10^{13}$ GeV, Yukawa range $\sim 10^{-27}$ cm) for SM matter, and via exact conformal decoupling for the dark-matter fermion ψ ; $\gamma = 1$ within the Cassini bound.

4. **RG stability:** ξ and λ_0 are stable under RG flow; β_ξ is driven at 1-loop by the ψ Yukawa ($\propto g^2$) and a direct quartic-curvature term ($\propto \lambda_0$, same sign), totaling $\Delta\xi \leq 10^{-6}$ at the fiducial $\xi = 11.1$ (quartic-dominated; across $\xi \in [1, 100]$ the running stays $\leq 10^{-5}$), a robust consequence of the small $\lambda_0 \sim 10^{-7}$ and $g \simeq 10^{-7}$ singlet embedding.

b. Level 2: Dark matter prediction (quantitative matching; normalization from the homogeneous mode computation, lattice cross-check open).

5. The dark matter is a Dirac fermion ψ whose abundance is matched to Ω_{DM} at $g \simeq 1.5 \times 10^{-7}$, i.e. $m_\psi \simeq 6.6 \times 10^{-3} H_{\text{inf}} \simeq 1.1 \times 10^{11}$ GeV, stable under the discrete $\mathbb{Z}_2^\psi$ symmetry (assuming an anomaly-free UV gauge completion) and the conformally vanishing $\chi\psi\bar{\psi}$ vertex, and produced gravitationally (non-adiabatic metric change at the end of inflation, in the Einstein frame). The absolute normalization is obtained from the exact homogeneous-background mode computation (transition + oscillating-condensate era, Appendix D0); a full Floquet/lattice analysis remains as the non-perturbative cross-check.

c. Level 3: Structural properties (not observational predictions).

6. **$\mathbb{Z}_2$ domain-wall thin-shell matching degeneracy:** $F(0) = F'(0) = 0$ renders the Israel–Lanczos conditions algebraically inconsistent. This is a mathematical property of $F = \xi\Phi^2$, observationally inert since $\mathbb{Z}_2$ walls are inflationarily diluted. The bubble-wall channel is not ruled out by the $F(0) = 0$ degeneracy, but its cosmological viability has not been verified (nucleation-rate calculation pending).
7. The 10^{110} hierarchy between the inflationary scale and dark energy is spanned by the conformal factor $e^{-4\varphi}$ but not dynamically explained, sharing the cosmological constant problem with all Λ CDM models.
8. The fine-structure-constant drift $\Delta\alpha/\alpha \leq 10^{-110}$ is > 90 orders below current sensitivity.

11 Discussion and outlook

1. **Inflationary initial conditions.** The $\chi \gg M_{\text{Pl}}$ condition is assumed, not derived. Whether a UV-complete theory naturally selects large-field Φ is an open problem.
2. **Dark matter: stability and relic abundance.** The dark matter is the Dirac fermion ψ , stable under the discrete $\mathbb{Z}_2^\psi$ symmetry (conditional on UV gauge completion) plus the conformally vanishing $\chi\psi\bar{\psi}$ vertex (Sec. V). Its relic abundance is set by gravitational particle production at the end of inflation—the non-adiabatic change of the Einstein-frame metric, which in the Jordan frame appears as the time-varying mass $m_\psi(t) = g|\Phi(t)|$ (a single physical process, not two additive channels). A quantitative Ω_ψ normalization is obtained from the exact homogeneous-background mode integration (Sec. V, Appendix D0): the abundance matching sits on the light branch at $m_\psi \simeq 6.6 \times 10^{-3} H_{\text{inf}} \simeq 1.1 \times 10^{11}$ GeV ($g \simeq 1.5 \times 10^{-7}$), the dominant residual uncertainty being the transition-shape systematic (factor $\simeq 1.6$) quoted there; a dedicated Floquet/lattice simulation remains the non-perturbative cross-check of the homogeneous treatment.

3. **Preheating energy partition.** The gravitational ψ production and the condensate decay through conformal-anomaly channel compete for the condensate energy. If the ψ channel is too efficient, an excessive fraction of the condensate energy is diverted into stable matter rather than radiation, which could suppress the reheating temperature. In the homogeneous treatment the split is already determined: ψ is a test field with $\rho_\psi/\rho_{\text{end}} \sim 10^{-19}$ at the matching point (Appendix D0), so the radiation/ ψ channels are independent and the partition is set by the anomaly decay; a lattice simulation would verify that this test-field hierarchy survives the full 3D preheating dynamics.
4. **Stochastic gravitational-wave background from preheating.** The oscillating condensate and any non-perturbative field inhomogeneities during preheating generically source a stochastic gravitational-wave background. For the present model, the extremely weak coupling of χ to SM fields ($g_{\chi ff} \sim m_f/M_{\text{Pl}} \leq 10^{-17}$) and the closure of the $\chi \rightarrow h\bar{h}$ channel suppress parametric resonance, so the inhomogeneous component of the condensate is expected to be small and the resulting GW background correspondingly weak. A rough estimate using the standard scaling $h^2\Omega_{\text{GW}} \sim 10^{-5}(H_{\text{reh}}/M_{\text{Pl}})^2$ with $H_{\text{reh}} \simeq \sqrt{\pi^2 g_*/90} T_{\text{reh}}^2/M_{\text{Pl}} \sim 1.4 \text{ GeV}$ (corresponding to $T_{\text{reh}} \sim 10^9 \text{ GeV}$) gives $h^2\Omega_{\text{GW}} \sim 3 \times 10^{-42}$, some 30 orders below the sensitivity of LISA ($\sim 10^{-12}$) and pulsar timing arrays ($\sim 10^{-10}$). A dedicated lattice computation of the GW spectrum is deferred to the same future preheating simulation that would verify the homogeneous Ω_ψ determination of Sec. V.
5. **Parameter status.** Φ is a gauge singlet, so the Standard-Model-Higgs β_ξ does not apply; λ_0 is fixed by $A_s = 2.1 \times 10^{-9}$ at the locked N (e.g. 6.70×10^{-8} at $N = 50$, $\xi = 11.1$), while $\xi \sim 11.1$ is compatible with $r < 0.036$. The primary predictions are the exact slow-roll pairs (n_s, r) at the locked N , not a single closed-form $r(N)$.
6. **Fifth-force screening and reheating closure.** Two structural features operate in distinct sectors: (a) the conformal identity $\Omega = \Phi/\Phi_0$ (a consequence of $F = \xi\Phi^2$) decouples the scalar from the dark-matter fermion ψ at tree level, closing the perturbative Yukawa reheating channel; (b) for Standard Model matter, whose masses arise via the Higgs mechanism, the residual χ coupling $\sim m_f/M_{\text{Pl}}$ is suppressed by the scalar's large mass $m_\chi \sim 10^{13} \text{ GeV}$ (Yukawa range $\sim 10^{-27} \text{ cm}$), satisfying the Cassini bound without a chameleon mechanism.
7. **$\mathbb{Z}_2$ domain-wall open problems.** The thin-shell matching degeneracy at $\Phi = 0$ shows that $\mathbb{Z}_2$ walls lie beyond the classical EFT applicability boundary. Open questions include: scalar hair configurations with $F(r_s) = 0$, the quantum-gravity meaning of $F(0) = 0$, and beyond-classical-EFT realizations of the $\mathbb{Z}_2$ channel. The bubble-wall channel ($F(\Phi_0) = M_{\text{Pl}}^2$) is not ruled out by the $F(0) = 0$ thin-shell degeneracy, but its cosmological viability requires a nucleation-rate analysis not performed here.
8. **EFT cutoff and UV completion.** From the Einstein-frame EFT perspective the cutoff is $\sim M_{\text{Pl}}$ and slow-roll predictions are self-consistent [3]; from the Jordan-frame perspective $\Phi_{\text{CMB}}/\Lambda_J \sim 30$ reflects a unitarity tension whose physical status in the Einstein frame is contested [13–15]. UV completion remains open.
9. **Swampland conjectures.** The de Sitter swampland conjecture [42] requires $|V'|/V \geq c \sim \mathcal{O}(1)$ or $V''/V \leq -c$ for any effective scalar potential derivable from

quantum gravity. The inflationary plateau of the present model ($V' \sim e^{-\varphi} \ll 1$ at $\varphi \gg 1$) is in tension with this conjecture, as are most single-field slow-roll models. Whether the conjecture survives as a rigorous constraint on low-energy EFTs remains unsettled [43]; we flag this as an open issue for any UV completion.

10. **Cosmological constant fine-tuning.** V_c is calibrated empirically; the 10^{110} hierarchy and the coincidence problem receive no dynamical explanation in this framework (see also Sec. VI).
11. **Observational indistinguishability from Λ CDM.** The dark-energy prediction $w = -1$ (up to $\leq 10^{-10}$ corrections) and the Λ CDM-identical structure-growth prediction (Sec. VI) imply that the late-time model is observationally indistinguishable from Λ CDM by any foreseeable survey. The testable content is therefore confined to the inflationary sector ($n_s, r, n_t, f_{\text{NL}}$) and the dark-matter relic abundance (normalization closed at the homogeneous level, Sec. V and Appendix D0; the 3D lattice cross-check remains); the dark-energy sector, while structurally unified with inflation, does not provide independent falsifiable signatures. Within the inflationary sector, moreover, $n_s(N)$ and $r(N)$ coincide numerically with the R^2 (Starobinsky) prediction at the same N (Sec. I), so an observed (n_s, r) pair fixes N without discriminating between the two models, and a LiteBIRD detection of $r > 0.01$ excludes the two together. The one genuinely non-degenerate relation of this framework is the dark-matter one, through the fixed relation (33): should a UV completion or any future non-gravitational effect ever fix g , Eq. (33) would translate it into a prediction for m_ψ at fixed ξ , and conversely—an internal consistency relation today, since ψ couples only gravitationally (Sec. V). This is the strongest discriminating statement the present version of the model supports, and it is exactly why the dark-matter normalization of Sec. V matters more here than in a model whose inflationary sector is already distinctive.
12. **Field-space structure at $\Phi = 0$.** The Jordan-frame kinetic metric $g_{\Phi\Phi} = \xi\Phi^2$ degenerates at $\Phi = 0$, rendering this point a boundary of the classical field space. Consequently, classical field configurations cannot cross $\Phi = 0$; the two $\mathbb{Z}_2$ vacua are topologically separated at the classical level. The thin-shell analysis of Sec. VIII confirms that the EFT itself signals its breakdown at $\Phi = 0$ through the matching-condition degeneracy, motivating a quantum-tunneling (instanton) description of any inter-vacuum transitions.
13. **Higher-order corrections to the dark-energy equation of state.** The $w = -1$ result (Sec. VI) follows from the quadratic expansion of $V_E(\varphi)$ near $\varphi = 0$. The quartic term $\sim \lambda_0 \Phi_0^4 \varphi^4 / M_{\text{Pl}}^4$ contributes a fractional correction $\Delta w/w \sim (\lambda_0/\xi^2)\varphi^2 \simeq 5.4 \times 10^{-10} \varphi^2$. For the frozen field value $\varphi \sim \delta\Phi/M_{\text{Pl}} \sim (H_0/m_\chi)^2 \sim 2 \times 10^{-111}$ (the Ricci driving of Sec. VI), this correction is $\sim 2 \times 10^{-231}$, utterly negligible. The harmonic approximation is therefore exact for all practical purposes.
14. **Running vacuum energy.** Mechanisms such as running vacuum energy ($V_c \propto H^2$) are not realized in this model, where V_c is a constant potential parameter. The cosmological constant problem—why $V_c \sim 10^{-47} \text{ GeV}^4$ —remains unsolved and is shared with Λ CDM (see fine-tuning discussion in Sec. VI).

15. **Hubble tension.** Because the late-time model is observationally indistinguishable from Λ CDM, it inherits the H_0 tension between Planck CMB ($H_0 \simeq 67.4$ km/s/Mpc) and local distance-ladder measurements ($H_0 \simeq 73.0$ km/s/Mpc) without providing a resolution. The model's testable content is confined to the inflationary sector ($n_s, r, n_t, f_{\text{NL}}$), where it is numerically degenerate with R^2 inflation, and to the dark-matter relic abundance; the dark-energy and late-expansion sectors offer no new leverage on the Hubble tension.
16. **Gravitational-wave speed and polarization.** In induced gravity with $F = \xi\Phi^2$, tensor perturbations propagate at the speed of light ($c_T = c$) in the Einstein frame at all epochs where Φ is frozen at its VEV ($\Phi = \Phi_0$, i.e. the entire late Universe), automatically satisfying the GW170817 constraint $|c_T/c - 1| \leq 10^{-15}$. During inflation, when Φ evolves, the time variation of the conformal factor $\Omega(t)$ could in principle modify c_T ; however, the deviation $\delta c_T/c \sim \dot{\Omega}/(H\Omega) \sim \mathcal{O}(\epsilon) \leq 10^{-2}$ affects only primordial tensor modes at horizon exit ($k \sim H_{\text{inf}}$) and does not propagate to LIGO-band frequencies today. Tensor perturbations retain the two standard GR polarization modes throughout; no scalar or vector polarizations are generated because the Einstein-frame graviton is minimally coupled to matter. The model therefore satisfies all existing GW speed and polarization constraints.
17. **Thermalization after reheating.** The reheating analysis assumes that decay products thermalize rapidly, establishing a well-defined temperature T_{reh} . At $T \sim 10^9$ GeV, the SM gauge interaction rate is $\Gamma_{\text{therm}} \sim \alpha_s^2 T \simeq 1.4 \times 10^6$ GeV (one-loop running from the Sec. IV anchor $\alpha_s(m_Z) = 0.1179$ gives $\alpha_s(10^9 \text{ GeV}) \simeq 0.038$), while the Hubble rate is $H \simeq \sqrt{\pi^2 g_*/90} T^2/M_{\text{Pl}} \sim 1.4$ GeV; the ratio $\Gamma_{\text{therm}}/H \sim 10^6$ still confirms near-instantaneous thermalization. Gravitationally produced particles are initially non-thermal, but their energy density is $\sim 10^{13}$ times smaller than the anomaly-driven radiation bath (Sec. IV), so they thermalize with the dominant component within a few interaction times, well before BBN.
18. **PPN parameter β .** The discussion of fifth-force screening in Sec. II E addresses the Eddington parameter γ ; the nonlinear PPN parameter β_{PPN} receives corrections from cubic self-interactions of χ . For $m_\chi \sim 10^{13}$ GeV, the effective $\beta_{\text{PPN}} - 1 \sim (g_{\chi\chi\chi}/M_{\text{Pl}})^2 (m_\chi r)^{-4} \sim 10^{-80}$ at Solar-System scales, far below the Cassini bound $\beta_{\text{PPN}} - 1 \leq 10^{-5}$. The remaining PPN parameters ($\xi, \alpha_1, \alpha_2, \alpha_3$) all vanish because the model possesses no preferred-frame or preferred-location effects in the Einstein frame.
19. **Additional consistency checks.** Several further consistency conditions are satisfied parametrically: (i) *Dark matter self-interactions:* ψ - ψ scattering via graviton exchange gives $\sigma_{\psi\psi}/m_\psi \sim G_N^2 m_\psi \sim 1 \times 10^{-69} \text{ cm}^2/\text{g}$, ~ 69 orders below the bullet-cluster bound ($\leq 1 \text{ cm}^2/\text{g}$); loop-induced scalar exchange is further suppressed by $g_{\chi\psi\psi}^{\text{eff}} \sim 10^{-11} g$. (ii) *Tremaine-Gunn bound:* the non-thermal ψ phase-space density $Q \equiv \rho/\sigma_v^3$ at production (with velocity dispersion $\sigma_v \sim H_{\text{inf}}/m_\psi$) is $Q \sim m_\psi^4 \sim 1.4 \times 10^{44} \text{ GeV}^4$, far above the dwarf-galaxy lower limit. (iii) *Graviton dark radiation:* relativistic gravitons produced during reheating contribute $\Delta N_{\text{eff}} \leq 10^{-5}$, negligible compared to Planck. (iv) *Entropy production:* irreversible; total entropy per comoving volume increases by $\sim 10^{13}$ during reheating. (v) *Neutrino sector:* χ - ν

effective coupling $\sim m_\nu/M_{\text{Pl}} \leq 10^{-27}$, no observable consequences. (vi) *Primordial magnetic fields*: anomaly-induced seed after a^{-2} redshift gives $B_0 \sim 10^{-25}$ G, far below blazar lower bounds. (vii) *Geodesic deviation*: Einstein-frame geodesic equation reproduces Yukawa suppression for SM fermions. (viii) *Gravitational production of ψ* : the abundance is obtained from the Bogoliubov coefficients of the mode equation across the end-of-inflation transition and the oscillating-condensate era, not from an instantaneous-pulse approximation; the homogeneous computation is validated by the convergence/unitarity/conformal-limit battery of Appendix D0, and a full Floquet/lattice treatment remains as the 3D cross-check (Discussion). (ix) *Particle production backreaction*: $\rho_{\text{part}} \leq 10^{-13} \rho_\chi$. (x) *Late-time scalar-photon coupling*: mixing angle $\sim 10^{-55}$, CMB distortions $\leq 10^{-60}$. (xi) *Ghost freedom*: Einstein-frame theory is GR plus a minimally coupled canonical scalar; kinetic terms positive-definite. (xii) *Cosmic strings from $U(1)_D$ breaking*: inflated away or never form within horizon. (xiii) *Jordan-frame PPN cross-check*: $|\gamma_J - 1| \leq 10^{-5}$. (xiv) *Spectral index running*: $\alpha_s \simeq -2/N^2 \sim -8 \times 10^{-4}$, consistent with Planck. (xv) *CMB B-mode spectrum*: peaks at $\ell \sim 80$ for $r \simeq 0.005$, within LiteBIRD reach. (xvi) *Domain-wall observational null*: $e^{-3N} \sim 10^{-66}$ renders all wall observables negligible.

12 Conclusions

We have constructed a scalar-tensor cosmology within pure induced gravity, where a single real scalar field Φ with non-minimal coupling $\xi\Phi^2 R$ and $\mathbb{Z}_2$ -symmetric potential $V_J(\Phi) = \lambda_0(\Phi^2 - \Phi_0^2)^2/4 + V_c$ simultaneously accounts for the Planck mass, inflation, dark matter, and dark energy. The framework is minimal in its field content—one scalar, one Dirac fermion, the Standard Model—and its key structural features follow unavoidably from the induced-gravity coupling $F = \xi\Phi^2$.

Rigorous results

(exact analytic derivations):

- The conformal transformation maps the Jordan-frame double-well potential to a Starobinsky-type plateau. Under the locked definition $N = \ln(a_{\text{end}}/a_*)$, exact potential slow-roll with A_s normalization gives (e.g. $\xi = 11.1$) $n_s \simeq 0.962$, $r \simeq 0.0043$ at $N = 50$; anomaly $T_{\text{reh}} \sim 10^9$ GeV corresponds to $N \simeq 50.7$.
- The conformal identity $\Omega = \Phi/\Phi_0$ —a direct consequence of $F = \xi\Phi^2$ —closes the tree-level $\chi\psi\bar{\psi}$ and $\chi \rightarrow h\bar{h}$ vertices exactly, forcing reheating through the conformal-anomaly channel.
- The discrete $\mathbb{Z}_2^\psi$ symmetry (under an assumed anomaly-free UV gauge completion), together with the vanishing $\chi\psi\bar{\psi}$ vertex, guarantees ψ stability; the condensate decays completely, leaving V_c as a frozen cosmological constant with $w = -1$ (up to residual corrections $\leq 10^{-10}$ bounded in App. D5).
- RG analysis shows $\beta_{\lambda_0} \propto \lambda_0^2 > 0$ and $\Delta\xi \leq 10^{-6}$, confirming vacuum stability and the absence of fine-tuned running.

Quantitative estimates

(parametric derivations with controlled uncertainties):

- N is defined as $N = \ln(a_{\text{end}}/a_*)$ with explicit matching assumptions (14). Planck 2σ selects $N \approx 45\text{--}56$ (the statistical n_s band; instantaneous reheating caps the physical window at $N_{\text{max}} \simeq 55.6$); anomaly-dominated $T_{\text{reh}} \sim 10^9$ GeV maps to $N \simeq 50.7$ in this convention (self-consistent T_{reh}^* for $N = 50$ is $\simeq 1.1 \times 10^8$ GeV).
- Throughout the Planck 2σ band, exact slow roll gives $r \leq 0.0052$ at $\xi = 11.1$; LiteBIRD $r > 0.01$ would exclude the class.
- The dark-matter mass is fixed by the abundance matching at $m_\psi \simeq 6.6 \times 10^{-3} H_{\text{inf}} \simeq 1.1 \times 10^{11}$ GeV ($g \simeq 1.5 \times 10^{-7}$) and follows from the gravitational-production mechanism; the absolute normalization is obtained from the exact homogeneous-background mode computation (Appendix D0), with a Floquet/lattice simulation as the remaining cross-check.

Open questions

(deferred to future work or UV completion):

- Inflationary initial conditions ($\chi \gg M_{\text{Pl}}$) and the measure of the slow-roll attractor.
- The 10^{110} hierarchy between V_0 and V_c (the cosmological constant problem, shared with Λ CDM).
- Non-perturbative (3D lattice) confirmation of the homogeneous preheating treatment: the absolute normalization of the dark-matter relic abundance is obtained at the homogeneous level (Sec. V, Appendix D0), and the preheating energy partition is fixed there by the test-field hierarchy $\rho_\psi/\rho_{\text{end}} \sim 10^{-19}$; what remains for the lattice is the 3D dynamics of fragmentation and resonance.
- UV completion: anomaly cancellation for $\mathbb{Z}_2^\psi$, EFT cutoff and unitarity, swampland constraints.
- Baryogenesis and the Hubble tension (inherited from Λ CDM without resolution).

The key observational test is r under the locked N definition: Planck 2σ allows $N \approx 45\text{--}56$ with $r \leq 0.0052$ (the statistical n_s band, capped at $N_{\text{max}} \simeq 55.6$ by instantaneous reheating; Table 1). A LiteBIRD-class measurement $r > 0.01$ at 95% CL would exclude the model class in this convention. The frozen $w_0 = -1$ prediction is shared with Λ CDM, so it discriminates neither model from the other; it is nonetheless a live exposure, since DESI DR2 already prefers $w_0 \neq -1$ at 3.1σ [36], and a confirmation would exclude this dark-energy sector (though not the inflationary one). The inflationary observables n_s , r , n_t and f_{NL} are the sharpest handles for *excluding* the model class, but at a given N they are numerically identical to the R^2 (Starobinsky) predictions and therefore do not discriminate between the two; the one non-degenerate relation of the framework is the dark-matter relation (33), whose observational content is fixed at the homogeneous level by the normalization of Sec. V ($g \simeq 1.5 \times 10^{-7}$, with the transition-shape systematic quoted there), pending only the non-perturbative cross-check.

Calibration vs. prediction (summary).

Rigid predictions of the action at chosen (ξ, N) are the inflationary functions $n_s(N)$ and $r(N)$ and the derived scales H_{inf} , m_χ , $U^{1/4} = (3M_{\text{Pl}}^2 H_{\text{inf}}^2)^{1/4}$. Calibrated or a-posteriori inputs are: Ω_Λ (hence V_c), λ_0 (from A_s), ξ (within the r bound), and the dark-matter Yukawa g (from Ω_{DM} ; matched at $g \simeq 1.5 \times 10^{-7}$ on the light branch by the exact homogeneous-background computation of Appendix D0, with the lattice cross-check open). Dark energy is structurally present in the action but observationally degenerate with Λ CDM; the hard falsification handle lives in the inflationary sector.

Appendix A Numerical verification

The full CMB normalization proceeds in four steps:

Step 1: Plateau height from the potential.

The Einstein-frame potential for $\chi \gg M_{\text{Pl}}$ ($\varphi \gg 1$) is

$$V_E(\chi) \simeq V_0 \left(1 - 2e^{-\beta\chi/M_{\text{Pl}}}\right), \quad V_0 \equiv \frac{\lambda_0 M_{\text{Pl}}^4}{4\xi^2}. \quad (\text{A1})$$

Step 2: Slow-roll parameters and e -folds.

With the potential slow-roll expressions of Eq. (22), the e -fold number between CMB horizon exit (χ_*) and the end of inflation ($\epsilon_V = 1$ at χ_{end}) is

$$N \equiv \int_{\chi_*}^{\chi_{\text{end}}} \frac{H}{d\chi/dt} d\chi \simeq \frac{1}{M_{\text{Pl}}^2} \int_{\chi_*}^{\chi_{\text{end}}} \frac{V_E}{V_E'} d\chi = \frac{e^{\beta\chi_*/M_{\text{Pl}}}}{2\beta^2} - \frac{1}{2\beta^2} - \frac{\sqrt{2}}{\beta}, \quad (\text{A2})$$

where the last two terms are subleading for $N \geq 50$.

Step 3: Scalar power spectrum.

The standard single-field slow-roll formula gives

$$A_s = \frac{V_E}{24\pi^2 M_{\text{Pl}}^4 \epsilon_V} \Big|_{\chi=\chi_*} \simeq \frac{V_0}{24\pi^2 M_{\text{Pl}}^4} \frac{1}{2\beta^2 e^{-2\beta\chi_*/M_{\text{Pl}}}} = \frac{\lambda_0}{192\pi^2 \xi^2 \beta^2} e^{2\beta\chi_*/M_{\text{Pl}}}. \quad (\text{A3})$$

Using $e^{\beta\chi_*/M_{\text{Pl}}} \simeq 2\beta^2 N$ from Step 2,

$$A_s \simeq \frac{\lambda_0 \beta^2 N^2}{48\pi^2 \xi^2} = \frac{\lambda_0 N^2}{12\pi^2 \xi^2 (6 + 1/\xi)}, \quad (\text{A4})$$

where we used $\beta^2 = 4/(6 + 1/\xi)$ so that $\beta^2/48 = 1/[12(6 + 1/\xi)]$. Equation (A4) is algebraically identical to the result obtained directly from $A_s = V_0/(24\pi^2 M_{\text{Pl}}^4 \epsilon_*)$ with $\epsilon_* = \beta_o^2/(8N^2)$, $\beta_o^2 = 6 + 1/\xi$.

Step 4: Solving for λ_0 .

Setting $A_s = 2.1 \times 10^{-9}$ (Planck 2018 [16]) and $N = 50$ gives the *large-field analytic* estimate

$$\lambda_0 = \frac{12\pi^2 \xi^2 (6 + 1/\xi) A_s}{N^2} \simeq 7.46 \times 10^{-8} \quad (\xi = 11.1, N = 50) \text{ (analytic; not the locked-} N \text{ table value).} \quad (\text{A5})$$

Note. The shortcut formula $\lambda_0 \simeq 48\pi^2 \xi^2 A_s / N^2$ (4.90×10^{-8} at $\xi = 11.1$, $N = 50$) omits the factor $(6 + 1/\xi)$ and is inconsistent with Eq. (A4). The large-field analytic estimate (A5) gives $\lambda_0 \simeq 7.46 \times 10^{-8}$; the **locked- N exact slow-roll inversion** used throughout the main text (Eqs. (15)–(16)) gives $\lambda_0 \simeq 6.70 \times 10^{-8}$ at $N = 50$ and $\simeq 6.45 \times 10^{-8}$ at $N = 51$. Table 1 and all fiducial scales quote the locked- N exact values.

The computational core is a one-dimensional root-find: at fixed N and ξ , λ_0 is obtained by requiring that the exact potential slow-roll amplitude A_s (Eqs. (15)–(16)) equals the observed 2.100×10^{-9} , with x_* determined by $N = \ln(a_{\text{end}}/a_*)$ and $\epsilon_V(x_{\text{end}}) = 1$. The closed form quoted below is the *analytic large-field estimate* (A5) and is *not* the locked- N value; the two differ because the analytic form neglects the $\mathcal{O}(1)$ correction from evaluating ϵ_V, η_V at the exact $x_*(N)$. The complete, executable implementation is `lock_n_convention.py` (together with `background_and_reheating.py`); a ten-line listing cannot faithfully reproduce the locked value. For orientation:

```
import numpy as np
from scipy.optimize import brentq
xi = 11.1; M_P = 2.435e18; As_target = 2.1e-9
beta = 2.0/np.sqrt(6+1/xi) # paper convention
beta_o2 = 6+1/xi # our convention
N_fid = 50
# Analytic large-field estimate ONLY (Eq. A5): gives lam0 = 7.465e-08, NOT the locked value
lam0 = 12*np.pi**2*xi**2*beta_o2*As_target/N_fid**2
# Locked-N value requires inverting As at fixed N (root-find); results:
# N=50 -> lam0=6.70e-08, ns=0.9616, r=0.00425
# N=51 -> lam0=6.45e-08, ns=0.9623, r=0.00409; T_reh*(N=51)=2.2e9 GeV (Table I)
# H = sqrt(lam0)*M_P/(2*np.sqrt(3)*xi)
```

Key results (locked N , $\xi = 11.1$).

At $N = 50$: $\lambda_0 = 6.70 \times 10^{-8}$, $n_s = 0.9616$, $r = 0.00425$, $V_0 \simeq 4.78 \times 10^{63} \text{ GeV}^4$, $H_{\text{inf}} \simeq 1.64 \times 10^{13} \text{ GeV}$, $m_\chi \simeq 3.25 \times 10^{13} \text{ GeV}$, $U^{1/4} \simeq 8.3 \times 10^{15} \text{ GeV}$. At $N = 51$ ($T_{\text{reh}}^* = 2.2 \times 10^9 \text{ GeV}$, Table 1): $\lambda_0 = 6.45 \times 10^{-8}$, $n_s = 0.9623$, $r = 0.00409$, $H_{\text{inf}} \simeq 1.61 \times 10^{13} \text{ GeV}$; the anomaly value $T_{\text{reh}} \sim 10^9 \text{ GeV}$ maps instead to the interpolated $N \simeq 50.7$ (Sec. III). In all cases $U^{1/4} = (3M_{\text{Pl}}^2 H_{\text{inf}}^2)^{1/4}$.

Appendix B Conformal decoupling: screening and reheating

The structural reason the scalar decouples from the dark-matter fermion ψ and the Yukawa reheating channel closes is the exact proportionality $\Omega = \Phi/\Phi_0$, which follows

from $F(\Phi) = \xi\Phi^2$ and is specific to the induced-gravity coupling. (Solar-system fifth-force bounds on Standard Model matter are instead met by the large scalar mass m_χ ; see Sec. II E.)

Matter coupling (ψ sector).

Under $\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}$, $\sqrt{-g} = \Omega^{-4} \sqrt{-\tilde{g}}$ and $\psi = \Omega^{3/2} \tilde{\psi}$. The Yukawa term transforms as

$$-g\Phi\bar{\psi}\psi\sqrt{-g} = -g\Phi_0\Omega\cdot\Omega^3\cdot\Omega^{-4}\tilde{\psi}\tilde{\psi}\sqrt{-\tilde{g}} = -g\Phi_0\tilde{\psi}\tilde{\psi}\sqrt{-\tilde{g}}, \quad (\text{B6})$$

giving an exact, χ -independent Einstein-frame mass $m_E = g\Phi_0$ and a vanishing tree-level $\chi\tilde{\psi}\tilde{\psi}$ vertex. Static ψ matter in this sector does not source χ , so $\gamma = 1$ exactly within the ψ sector (observationally inert for solar-system tests, which probe SM matter; decisive for reheating and dark matter).

Spin-connection cancellation.

Under conformal transformation, the vierbein and Dirac matrices transform as $\tilde{e}_\mu^a = \Omega e_\mu^a$, $\tilde{\gamma}^\mu = \tilde{e}_a^\mu \gamma^a = \Omega^{-1} \gamma^\mu$, and the spinor rescales as $\psi = \Omega^{3/2} \tilde{\psi}$. The massless Dirac kinetic term becomes

$$\begin{aligned} S_D &= \int d^4x \sqrt{-g} \bar{\psi} i \gamma^\mu (\partial_\mu + \frac{1}{4} \omega_\mu^{ab} [e] \gamma_{ab}) \psi \\ &= \int d^4x \sqrt{-\tilde{g}} \tilde{\bar{\psi}} i \tilde{\gamma}^\mu \left[\partial_\mu \tilde{\psi} + \frac{3}{2} (\partial_\mu \ln \Omega) \tilde{\psi} + \frac{1}{4} \omega_\mu^{ab} [e] \gamma_{ab} \tilde{\psi} \right]. \end{aligned} \quad (\text{B7})$$

Using $\omega_\mu^{ab} [e] = \omega_\mu^{ab} [\tilde{e}] - (e_\mu^a e_\nu^b - e_\mu^b e_\nu^a) \partial_\nu (\ln \Omega)$ and the identity $\gamma^a \gamma_{ab} = 3\gamma_b$ in $d = 4$, the spin-connection contribution becomes

$$\frac{1}{4} \tilde{\gamma}^\mu \omega_\mu^{ab} [e] \gamma_{ab} = \frac{1}{4} \tilde{\gamma}^\mu \omega_\mu^{ab} [\tilde{e}] \gamma_{ab} - \frac{3}{2} \tilde{\gamma}^\nu \partial_\nu (\ln \Omega), \quad (\text{B8})$$

which exactly cancels the $+\frac{3}{2}(\partial_\mu \ln \Omega)$ rescaling term, leaving $S_D = \int d^4x \sqrt{-\tilde{g}} \tilde{\bar{\psi}} i \tilde{\gamma}^\mu \tilde{\nabla}_\mu \tilde{\psi}$ with no derivative coupling to χ .

Reheating.

The vanishing vertex means no perturbative $\chi \rightarrow \tilde{\psi}\tilde{\psi}$ decay, for any g . For $m_\psi \gtrsim m_\chi/2$ the decay is additionally kinematically forbidden; at the abundance-matched value $g \simeq 1.5 \times 10^{-7}$ of Sec. V the kinematic closure no longer holds but the vertex closure still does. The time-varying Jordan-frame mass $m_\psi = g|\Phi|$ is adiabatic for $g \gtrsim 10^{-4}$ and non-adiabatic at the dark-matter value $g \simeq 1.5 \times 10^{-7}$; this is the Jordan-frame image of the Einstein-frame gravitational production that sources the dark matter of Sec. V (a single physical process, not an additive channel), and it does not use the perturbative vertex. Reheating proceeds through conformal-anomaly production together with condensate decay (the gravitational pulse provides only a seed bath and cannot reheat on its own); under the locked N definition this allows $N \simeq 45\text{--}56$ by Planck 2σ (capped at $N_{\text{max}} \simeq 55.6$; Sec. III), with anomaly $T_{\text{reh}} \sim 10^9 \text{ GeV}$ at $N \simeq 50.7$ (Sec. III, Table 1).

Appendix C Key derivations

1. Variational derivation of Eq. (3).

$\delta S/\delta g_{\mu\nu} = 0$ with $\delta(\sqrt{-g}\frac{1}{2}FR) = \frac{1}{2}F\sqrt{-g}G_{\mu\nu}\delta g^{\mu\nu} - \frac{1}{2}\sqrt{-g}(\nabla_\mu\nabla_\nu F - g_{\mu\nu}\square F)\delta g^{\mu\nu}$. Adding kinetic and potential variations yields Eq. (3).

2. Conformal transformation.

$\Omega^2 = \xi\Phi^2/M_{\text{Pl}}^2$, $\tilde{R} = \Omega^{-2}(R - 6\Omega^{-1}\square\Omega)$, producing $d\chi/d\varphi = M_{\text{Pl}}\sqrt{6+1/\xi}$ and $V_E = V_J/\Omega^4$ (Eq. (8)).

3. Linearized field equation.

Linearizing $\square\Phi - V'(\Phi) + \xi\Phi R = 0$ with $\Phi = \Phi_0 + \delta\Phi$ and $V''(\Phi_0) = 2\lambda_0\Phi_0^2$ yields Eq. (37); the quasi-static limit ($\delta\ddot{\Phi} \ll H\delta\dot{\Phi} \ll m_{\text{eff}}^2\delta\Phi$) gives Eq. (38). For $m_\chi \gg H$ at all epochs after reheating (the ratio m_χ/H stays above $\sim 5 \times 10^{14}$ at the model's own reheating temperature and reaches 2.3×10^{55} today), the quasi-static condition is satisfied to extreme precision, justifying the neglect of time-derivative terms.

4. Junction-condition integration.

Integrating Eq. (3) across the shell gives the scalar-tensor Lanczos condition (48). At $\Phi_s = 0$, $F(0) = F'(0) = 0$ and the condition degenerates.

5. Divergent σ_E limit.

$\sigma_E = \int \sqrt{2V_E}(d\chi/d\varphi)d\varphi \sim \int e^{-2\varphi}d\varphi$ diverges at $\varphi \rightarrow -\infty$ ($\Phi = 0$), a conformal-mapping artifact.

Appendix D Post-inflationary field dynamics and the dark-energy equation of state

This appendix supplies the rigorous derivation of the dark-energy results quoted in Sec. VI: the harmonic-minimum approximation, the coherent-oscillation and condensate-decay regimes, the overclosure bound, and the complete residual-motion budget (three sources, of which the residual-condensate tail is not manifest in the main text).

0. Numerical validation of the end-of-inflation state.

The quantities used in Secs. III–V are obtained by integrating the exact Klein–Gordon equation in the Einstein frame, $\ddot{\chi} + 3H\dot{\chi} + V_E'(\chi) = 0$ with $3M_{\text{Pl}}^2H^2 = \frac{1}{2}\dot{\chi}^2 + V_E$, rather than by evaluating slow-roll closed forms. Two conventions matter and are stated explicitly here. (i) In e -fold variables the Friedmann constraint reads $H^2 = V_E/(3 - p^2/2)$ with $p \equiv d\chi/dN$, which degenerates as $V_E \rightarrow 0$; we therefore integrate the e -fold system only over the slow-roll regime and switch to cosmic time for the oscillatory phase. (ii) The end of inflation is defined by $\epsilon_V = 1$, where slow roll has already broken down; the slow-roll velocity $\dot{\chi} = -V_E'/(3H)$ must *not* be imposed there, since it would force $K_{\text{end}} = V_{\text{end}}/3$ and $\epsilon_H = 3/4$ by construction. Integrating instead from

$\epsilon_V = 0.05$ up to $\epsilon_V = 1$ gives $x_{\text{end}} = 0.76367$, $\epsilon_H(x_{\text{end}}) = 0.4988$, $V_{\text{end}}/V_0 = 0.28520$, $K_{\text{end}}/V_{\text{end}} = 0.1994$, and $\rho_{\text{end}}/V_{\text{end}} = 1.1994$. These reproduce the locked- N fiducial values quoted in Sec. III and give the barrier ratio $K_{\text{end}}/\Delta V_J \simeq 0.057$ of Sec. III. The same framework extends through the oscillating-condensate era: the mode equation is integrated on the exact homogeneous background (real slow-roll descent, numerically resolved condensate oscillations with $\langle w \rangle \simeq -0.02$ over the first e -folds, and an analytic matter-domination extension), with controlled backgrounds isolating the two systematics. The oscillating condensate raises the production by $R \equiv n_{\text{exact}}/n_{\text{avg}} \simeq 1.24$ at the matching point, while replacing the instantaneous-splice transition by the real descent lowers the transition-only production baseline by a factor $\simeq 2.7$ (a factor $\simeq 1.6$ in the matched g , since $g \propto n_\psi^{-1/2}$ on the light branch); the net effect moves the matched coupling from $g \simeq 1.0 \times 10^{-7}$ to $g \simeq 1.5 \times 10^{-7}$ ($m_\psi \simeq 1.1 \times 10^{11}$ GeV at $\Omega_\psi = 0.265$, the model's own $T_{\text{reh}} = 2.1 \times 10^8$ GeV). The computation is validated against the published transition-only spectrum (reproduced at the 10^{-9} level on the shared $p = 2$ background), the massless/conformal limit (exactly zero production), unitarity ($||\alpha|^2 + |\beta|^2 - 1| \lesssim 10^{-12}$), and a convergence battery in the phase step, k grid, starting e -folds, freeze criterion and integration endpoint. The same integration supplies the reheating ceiling used in Sec. III: $T_{\text{reh}}^{\text{inst}}(N) = (30\rho_{\text{end}}(N)/\pi^2 g_*)^{1/4}$ decreases only as $N^{-1/2}$, whereas the tabulated $T_{\text{reh}}^*(N)$ required by cosmological matching grows as $e^{3.05N}$; the two intersect at $N_{\text{max}} = 55.6$ (or 55.9 if the naive ceiling $V_{\text{end}}^{1/4}$ is used instead). This is the reason the n_s -allowed band is quoted as $N \approx 45\text{--}56$: with $N_{\text{max}} = 55.6$, the points $N = 56, 57$ and 58 would require a reheating temperature above instantaneous reheating. The numerical results quoted above and in Secs. III–VI were obtained with the analysis scripts accompanying this work: `background_and_reheating.py` (exact Klein–Gordon integration, reheating channels and the N window), `psi_production_bogoliubov.py` (de Sitter Bogoliubov index and the abundance matching), `psi_abundance_oscillating.py` (mode-equation abundance across the end-of-inflation transition), `dm_gap_closure_test.py` (abundance matching on the light branch with the exact mode-equation spectrum across the end-of-inflation transition, and the free-streaming check), `psi_mode_oscillating.py` (exact-background mode integration through the oscillating-condensate era: controlled backgrounds, validation battery and the final matching of Sec. V), `treh_error_band.py` (propagation of the reheating-temperature uncertainty to (n_s, r) and to the dark-matter matching window), `residual_quintessence.py` (two-fluid Boltzmann integration and the Δw budget), and `lock_n_convention.py` (the locked- N grid behind Table 1 and the figures), on the shared constants and the independent action-level derivation of `cosmo_model.py` and `derive_from_action.py` and on the closed forms behind the order-of-magnitude claims in `order_estimates.py`, together with `verify_numerics.py`, `consistency_checks.py`, `audit_tex_numbers.py` and `audit_readme_numbers.py` (audits that recompute every quantitative claim printed here from the current scripts, check every table cell against the script that produces it, and scan for superseded values). All are publicly available at <https://github.com/huyamingc/induced-gravity-cosmology>, together with the L^AT_EX source of this manuscript and a single entry point, `scripts/run_all.py`, that regenerates the numerical results and figures quoted in this work.

1. Equation of motion near the minimum.

After conformal transformation to the Einstein frame, the canonical field χ obeys

$$\ddot{\chi} + 3H\dot{\chi} + V'_E(\chi) = 0, \quad (\text{D9})$$

with

$$V_E(\chi) = V_c + \frac{1}{2}m_\chi^2\chi^2 + \mathcal{O}\left(\frac{m_\chi^2\chi^3}{M_{\text{Pl}}}\right), \quad (\text{D10})$$

where $m_\chi^2 \equiv 2\lambda_0 M_{\text{Pl}}^2/[\xi^2(6+1/\xi)] \approx (3.25 \times 10^{13} \text{ GeV})^2$ at locked $N = 50$ (Eq. (12)). The cubic term is *not* negligible at the largest post-inflationary amplitudes. Writing $V_E = V_0(1 - e^{-x})^2$ with $x \equiv \beta\chi/M_{\text{Pl}}$ and expanding about $x = 0$ gives $V_E = V_0[x^2 - x^3 + \mathcal{O}(x^4)]$, so the cubic-to-quadratic ratio is $\beta\chi/M_{\text{Pl}}$ —equal to $\beta \simeq 0.81$ at $\chi = M_{\text{Pl}}$, and therefore $\mathcal{O}(1)$ rather than $\mathcal{O}(\lambda_0^{1/2})$. The cubic coefficient is

$$c_3 = -\frac{V_0\beta^3}{M_{\text{Pl}}^3} = -\frac{\beta m_\chi^2}{2M_{\text{Pl}}} \simeq -1.8 \times 10^8 \text{ GeV}, \quad (\text{D11})$$

in the convention $V \supset c_3\chi^3$. The potential is thus effectively harmonic only for $\chi \ll M_{\text{Pl}}/\beta \simeq 1.2 M_{\text{Pl}}$; at the end of inflation, $\chi_{\text{end}} \simeq 0.94 M_{\text{Pl}}$, the anharmonicity is $\mathcal{O}(1)$. This is why the post-inflationary equation of state is not exactly $w = 0$ during the first oscillations—the exact integration of Sec. D0 gives $\langle w \rangle \simeq -0.02$ over the first e -folds, approaching zero thereafter—and it is the origin of the $\mathcal{O}(1)$ uncertainty in the mapping between N and T_{reh} noted in Sec. III. All numerical results in this work use the exact potential V_E , not the harmonic approximation. The ratio $m_\chi/H_0 \approx 2.3 \times 10^{55}$ implies that any displacement from $\chi = 0$ oscillates with frequency $\gg H$ and is damped to zero within $\sim m_\chi^{-1}$.

2. Coherent oscillation and the dust redshift law.

Immediately after the end of slow-roll ($\chi \sim M_{\text{Pl}}$, $\epsilon \sim 1$), the field begins coherent oscillations around $\chi = 0$. For $H \ll m_\chi$, Eq. (D9) reduces to a damped harmonic oscillator with WKB solution

$$\chi(t) = \chi_0(t) \cos\left(\int^t m_\chi dt' + \theta_0\right), \quad \chi_0(a) = \chi_{\text{end}} \left(\frac{a}{a_{\text{end}}}\right)^{-3/2}, \quad (\text{D12})$$

which follows from the conserved adiabatic invariant $I \equiv \oint p dq = \pi m_\chi \chi_0^2 a^3$. The energy density of the oscillating condensate therefore redshifts as pressureless dust,

$$\rho_{\text{cond}}(a) = \frac{1}{2}m_\chi^2\chi_0^2(a) \propto a^{-3}, \quad (\text{D13})$$

which is the rigorous origin of the $\rho \propto a^{-3}$ behavior quoted in Sec. IV.

3. Condensate decay: the overclosure bound.

The condensate must transfer its energy to SM radiation with high efficiency to avoid overclosing the universe. The observational constraint is

$$\rho_{\text{cond}}(a_{\text{reh}}) < \rho_{\text{DM}}^{(0)} \left(\frac{a_0}{a_{\text{reh}}} \right)^{-3} = \rho_{\text{DM}}^{(0)} \left(\frac{T_{\text{reh}}}{T_0} \right)^3, \quad (\text{D14})$$

where $\rho_{\text{DM}}^{(0)} = \Omega_{\text{DM}} \rho_{\text{crit}}^{(0)} \approx 9.7 \times 10^{-48} \text{ GeV}^4$ and $T_0 \approx 2.36 \times 10^{-13} \text{ GeV}$. For $T_{\text{reh}} \in [10^{-2}, 10^{15}] \text{ GeV}$ this gives $\rho_{\text{cond}}(a_{\text{reh}}) < (5.2 \times 10^{-16} - 5.2 \times 10^{35}) \text{ GeV}^4$. Comparing with the initial condensate energy density $\rho_{\text{cond}}(a_{\text{end}}) \sim \frac{1}{2} m_\chi^2 M_{\text{Pl}}^2 \simeq 3 \times 10^{63} \text{ GeV}^4$, the required suppression factor is $\sim 4 \times 10^{27} - 4 \times 10^{78}$ (the upper end corresponding to T_{reh} at the BBN floor). This suppression is supplied by the decay of the condensate (channel (iv) of Sec. IV): because $\Gamma t_0 \gg 1$ for any $T_{\text{reh}} \gtrsim 10^{-2} \text{ GeV}$, the residual condensate at a_0 is strictly zero (D5c), and the locked N is confined to the Planck-compatible band (anomaly $T_{\text{reh}} \sim 10^9 \text{ GeV} \Rightarrow N \simeq 50.7$).

4. Freezing at the minimum.

After reheating, the field sits at $\chi \approx 0$ with precision set by the residual amplitude. Since $m_\chi \gg H$ at all later epochs (including today, where $m_\chi/H_0 \sim 10^{55}$), any small residual oscillation is damped to $\chi = 0$ within

$$\sim m_\chi^{-1} \sim 2 \times 10^{-14} \text{ GeV}^{-1} \sim 10^{-38} \text{ s}, \quad (\text{D15})$$

effectively instantaneous compared to the Hubble time. With $\chi \equiv 0$ to extraordinary precision, the Einstein-frame scalar energy-momentum tensor reduces to $T_{\mu\nu}^{(\chi)} \rightarrow -V_c g_{\mu\nu}$, the energy-momentum tensor of a cosmological constant; the equation of state $w_0 = -1$ for all z then follows directly (Eq. (35)).

5. Complete residual-motion budget.

Sec. VI quotes the residual-motion bound $\Delta w \ll 10^{-3}$. Here we bound all three independent sources of residual χ motion that could perturb the frozen minimum.

a. Ricci driving. In the Jordan frame, the linearized equation (Eq. (37)) has a source term $\xi \Phi_0 R$ that displaces Φ from Φ_0 :

$$\delta \Phi_{\text{qs}} \approx \frac{\xi \Phi_0 R}{2 \lambda_0 \Phi_0^2}. \quad (\text{D16})$$

During matter domination $R \approx -3H^2(1-3w) \approx +3H^2$ for $w = 0$, so

$$\frac{\delta \Phi_{\text{qs}}}{\Phi_0} \sim \frac{\xi H^2}{\lambda_0 M_{\text{Pl}}^2} \sim \left(\frac{H}{m_\chi} \right)^2. \quad (\text{D17})$$

At $z = 0$ ($H = H_0$): $\delta\Phi/\Phi_0 \sim 10^{-111}$. Varying Φ produces kinetic energy $K \sim \frac{1}{2}\dot{\delta\Phi}^2 \sim \frac{1}{2}(\delta\Phi \cdot H)^2$, giving

$$\Delta w_{\text{Ricci}} \equiv \frac{K}{V_c} \sim \left(\frac{H_0}{m_\chi}\right)^2 \sim 2 \times 10^{-111}, \quad (\text{D18})$$

10^{108} orders below the DESI 3σ sensitivity [35].

b. Quantum fluctuations. For a massive scalar in de Sitter space with $m \gg H$, $\langle\chi^2\rangle \approx 3H^4/(8\pi^2 m_\chi^2)$. The corresponding energy density is $\rho_{\text{quant}} \sim 3H_0^4/(16\pi^2) \ll V_c$, yielding

$$\Delta w_{\text{quant}} \leq H_0^4/V_c \sim 2 \times 10^{-121}. \quad (\text{D19})$$

c. Residual condensate tail. This third source—not manifest in the main-text summary of Sec. VI—is the residual condensate that survives reheating. The condensate is tracked by the coupled two-fluid system $d(\rho_{\text{cond}}a^3)/dN = -(\Gamma/H)\rho_{\text{cond}}a^3$ and $d(\rho_{\text{rad}}a^4)/dN = +(\Gamma/H)\rho_{\text{cond}}a^4$, integrated from a_{end} to a_0 with $H^2 = (\rho_{\text{cond}} + \rho_{\text{rad}})/(3M_{\text{Pl}}^2)$ and an exponential integrator for the decay term (the ratio Γ/H reaches $\sim 10^{30}$ at late times). The result is that for every $\Gamma \neq 0$ the residual today underflows to zero: the decay-completeness condition $\Gamma t_0 \gg 1$ holds by ≥ 20 orders of magnitude across $T_{\text{reh}} \in [10^{-2}, 10^{15}]$ GeV. The overclosure bound is therefore not a constraint that happens to be met but a statement about what the residual *would* be for a stable condensate; that hypothetical value,

$$\rho_{\text{cond}}(a_0)|_{\Gamma=0} = \rho_{\text{end}}(a_{\text{end}}/a_0)^3 \simeq 1.65 \times 10^{-29} \text{ GeV}^4, \quad (\text{D20})$$

exceeds $\rho_{\text{DM}}^{(0)}$ by $\sim 10^{18}$ and would overclose the universe. With the decay included the upper bound

$$\rho_{\text{cond}}(a_0) < \rho_{\text{DM}}^{(0)} \approx 9.7 \times 10^{-48} \text{ GeV}^4, \quad (\text{D21})$$

giving the constraint $\rho_{\text{cond}}(a_0)/V_c \leq \Omega_{\text{DM}}/\Omega_\Lambda \sim 0.39$. This is a *requirement*, not a computed value: an absolutely stable condensate would give $\rho_{\text{cond}}(a_0)/V_c \simeq 7 \times 10^{17}$ and would overclose the universe by $\sim 10^{18}$. The requirement is met because the condensate decays: the decay-completeness condition $\Gamma t_0 \gg 1$ holds by ≥ 20 orders of magnitude for any $T_{\text{reh}} \gtrsim 10^{-2}$ GeV, so the residual at a_0 is strictly zero (see App. D5c). This bounds the residual condensate’s contribution to the dark-energy equation of state; combined with the rapid damping (Eq. (D15)) it confirms that the condensate is cosmologically irrelevant at late times, as asserted in Sec. VI.

6. Summary.

The complete dark-energy sector is captured by

$$V_E(0) = V_c, \quad m_\chi^2 = \frac{2\lambda_0 M_{\text{Pl}}^2}{\xi^2(6 + 1/\xi)}, \quad w_0 = -1, \quad (\text{D22})$$

with

$$\begin{aligned} |\Delta w| &< (H_0/m_\chi)^2 \sim 2 \times 10^{-111} \text{ (Ricci)} \\ &+ \leq 2 \times 10^{-121} \text{ (quantum)} \\ &+ 0 \text{ (condensate tail; } \Gamma t_0 \gg 1), \end{aligned} \quad (\text{D23})$$

the third line vanishes because $\Gamma t_0 \gg 1$ for any $T_{\text{reh}} \gtrsim 10^{-2} \text{ GeV}$ (see App. D5c), so the condensate tail contributes to Ω_m but not to Δw_{DE} ; and $\Delta \mathcal{A}/\mathcal{A} = +\xi R/(\lambda_0 \Phi_0^2)$.

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- **Data availability:** This manuscript has no associated data; it is a theoretical study and all results are derived analytically or from the numerical integrations described in the appendices. The analysis scripts used to produce every numerical statement in this work, together with the manuscript source and the compiled PDF, are publicly available at <https://github.com/huyamingc/induced-gravity-cosmology>.
- **Code availability:** Analysis scripts are publicly available at <https://github.com/huyamingc/induced-gravity-cosmology>; `scripts/run_all.py` runs the complete verification suite.
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